

China Semiconductors

China Smart Driving tracker (2Q26): L2++ penetration rebounded to record high at 26%; Horizon takes the lead in L2+



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This proprietary tracker analyzes China's smart driving chip industry using raw data from Gasgoo. The dataset reflects vehicles sold to end users based on auto insurance registrations. It is important to note that there is typically a 1–2 quarters of lead time between smart driving SoC vendors selling chips to OEM/Tier-1 and the final vehicle sales. Despite this lag, these figures provide an effective indicator of industry trends and competitive dynamics over the medium to long term.

NOA penetration rebounded sharply to 43% in 2Q26 following the temporary weakness in 1Q26 at 32%. This is mainly driven by L2++ reached a new record high at 26%, up 7 ppt QoQ. L2++ penetration among Explorers and Legacy OEMs remain low at ~5% and ~40% respectively, vs. 80%+ by Pioneers. We expect Explorers OEMs to ramp up L2++ in the near term, with Legacy OEMs to follow after that. At the same time L2+ penetration also rebounded to 16%, up 3 ppt QoQ, but is only back to the level similar to 4Q26. Basic L1-L2 continue to shrink to 37%, -6 ppt QoQ, same for None&LO shrink to 21%, -4 ppt QoQ.

Horizon Robotics: Leader in L2+, but penetration in L2++ weaker than expected. 2Q26 also marked a milestone for Horizon, which surpassed NVIDIA for the first time to become the largest supplier in the L2+ segment by vehicle installations. However, Horizon's progress in L2++ was slower than initial expectation, with market share declining sequentially from 5% to 3% amid intensifying competition from Explorers' in-house solutions and emerging 3rd-party vendors such as Qualcomm. We believe this is mainly due to the car models that deployed HSD didn't capture strong sales, thus the most important variable for Horizon over the next few quarters remains its collaboration with BYD for HSD. If Horizon delivers its management guidance, we estimate Horizon has the potential to increase its L2+ market share from the current 32% to 55-60%; in L2++, securing a modest allocation within BYD's growing Urban NOA wallet could nearly triple Horizon's current 3% market share, even before considering broader TAM expansion or incremental wallet-share gains at other OEMs. From an investment perspective, we view clearer disclosure around Horizon's collaboration with BYD as the most key upside over 2H26.

For the other vendors, Black Sesame and Nvidia further lose share, while momentum for Qualcomm is strong thanks to Leapmotor strong sales. Black Sesame further share loss in L2+, still no OEM design win for L2++. New installation volumes of Black Sesame L2+ chips declined further, amid the absence of new OEM customer acquisitions and continued wallet-share losses at Geely. **NVIDIA** continued to lose share in both L2+ and L2++. In L2+, NV was overtaken by Horizon for the first time, while in L2++ its market leadership is increasingly challenged by both OEM in-house solutions and emerging 3rd-party suppliers.

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Qualcomm delivered the fastest growth from a low base. The deployment of SA8797P (600+TOPS) officially places the company among the leading L2++ SoC platforms, and its deployment across full vehicle lineups of Leapmotor has enabled its rapid ramp-up.

At the OEM level, **BYD's focus is increasingly shifting from L2+ to L2++**, positioning the company as the largest source of incremental Urban NOA demand over the next 12 months. BYD's supplier strategy is also evolving from the NVIDIA-centric approach to a multi-source ecosystem consisting of NVIDIA, Horizon, and in-house chips. **Geely continues to shift L2+ wallet share from Black Sesame to Horizon**, while whether it eventually employs other vendors beyond NVIDIA for L2++ solution remains an important question to monitor.

Chery added a 3rd J6P-equipped model in 2Q26, lifting Horizon to the No.2 wallet-share position within its L2++ platforms.

LiDAR penetration rebounded QoQ, with Hesai maintaining market leadership.

LiDAR is seeing rapid adoption in China, with penetration rising to 20% in 2Q26 (vs. ~15% in 1Q26). Hesai continued to lead the market with 47% share in 2Q26, followed by RoboSense (23%), Huawei (21%), and Seyond (9%). In China, with over 80% of Urban NOA-equipped models adopting LiDAR, rising Urban NOA penetration should drive further LiDAR adoption. As the global leader in LiDAR, Hesai is well positioned to capture this structural growth opportunity.

BERNSTEIN TICKER TABLE

			10 Aug 2026		TTM	Reported EPS				Reported P/E (x)		
			Closing	Price	Rel.							
Ticker	Rating	Cur	Price	Target	Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
9660.HK (Horizon Robotics)	O	HKD	5.10	10.00	(61.4)%	RMB	(0.81)	(0.25)	(0.11)	13.3	8.5	5.1
2533.HK (Black Sesame)	U	HKD	11.65	16.00	(64.5)%	RMB	(2.37)	(2.03)	(1.85)	7.8	4.5	3.2
HSAL.US (Hesai)	O	USD	19.22	30.00	(45.7)%	CNY	2.92	3.31	5.27	44.4	39.2	24.6
2525.HK (Hesai)	O	HKD	18.46	30.00	NA	CNY	2.92	3.31	5.27	5.4	4.8	3.0
9866.HK (NIO)	M	HKD	37.96	47.00	(29.1)%	CNY	(6.85)	(1.62)	(0.49)	(4.8)	(20.2)	(66.4)
NIO (NIO)	M	USD	4.74	6.00	(24.4)%	CNY	(6.85)	(1.62)	(0.49)	(4.7)	(19.8)	(65.0)
LI (Li Auto)	M	USD	12.95	15.50	(70.1)%	CNY	1.13	(4.88)	2.27	77.2	(17.9)	38.5
2015.HK (Li Auto)	M	HKD	50.10	61.00	(78.7)%	CNY	0.57	(2.44)	1.13	76.2	(17.7)	38.0
9868.HK (XPeng)	M	HKD	47.12	78.00	(71.3)%	CNY	(0.60)	(1.02)	0.23	(67.7)	(39.9)	179.4
XPEV (XPeng)	M	USD	12.13	20.00	(62.3)%	CNY	(1.20)	(2.03)	0.45	(68.4)	(40.3)	181.2
NVDA (NVIDIA)	O	USD	223.96	315.00	0.2%	USD	4.77	9.19	12.52	46.9	24.4	17.9
QCOM (Qualcomm)	M	USD	167.86	165.00	(8.6)%	USD	12.03	10.51	9.57	14.0	16.0	17.5
ASIAx			1,914.02									
SPX			7,757.64									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

NVDA, QCOM estimate is Adjusted EPS; 9660.HK, 2533.HK valuation is EV/Sales (x); NVDA, QCOM valuation is Adjusted P/E (x); NVDA base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Horizon Robotics (Outperform, PT HKD 10): Horizon Robotics is the leading provider of smart driving chips in China. They offer distinctive hardware-software integrated ADAS solutions, combining SoC and algorithms for L2, L2+, and L2++ ADAS systems. This integrated approach allows them to deliver high-performance chips/software at competitive prices, helping them capture market share. Additionally, a strong balance sheet enables Horizon to invest significantly in R&D, ensuring they maintain their technological leadership in the smart driving SoC, and in the future develop chips for robotics & expand globally through JV.

In terms of valuation, we set our 12-month price target at HKD 10. Our valuation is based on a 10.3x Enterprise value-to-sales multiple applied to 2BF (2H27–1H28) revenue of RMB 11.5 billion.

Black Sesame (Underperform, PT HKD 16): Black Sesame is a provider of smart driving chips in China. They focus primarily on L2+ SoC and is expanding to L2++, however lacking software expertise limits their ability to provide user-friendly chips, slowing their customer acquisition compared to Horizon. Without sufficient scale, the heavy R&D burden suggests they will need to raise capital frequently, which could dilute shareholder value.

For Black Sesame, we set our 12-month TP at HKD 16. Our valuation is based on a 4.0x Enterprise value-to-sales multiple applied to 2BF (2H27–1H28) revenue of RMB 2.3 billion. The target EV/sales multiple reflects our concerns regarding the slow pace of design wins and ongoing cash flow pressure from R&D burden.

NVDA (Outperform, \$315.00): The datacenter opportunity is enormous, and still early, with material upside still possible.

QCOM (Market-Perform, \$165.00): The diversification story is real but a number of near-term headwinds (smartphone memory dynamics, margins, AAPL etc) need to be navigated first.

Hesai (Outperform, PT US\$ 30 / HK\$ 30): Hesai is a global leader in LiDAR, well positioned to benefit from the rapid adoption of intelligent vehicles and the long-term growth of emerging robotics. We adjust our target price for Hesai's H-shares from HK \$238 to HK\$30 to reflect the recent 8-for-1 stock split, with no changes to our estimates.

For the Chinese EV names, we rate **NIO (PT US\$6/HK\$47)**, **XPeng (PT US\$20/HK\$78)**, and **Li Auto (PT US\$15.5/HK\$61) Market-Perform**.

DETAILS

Chinese auto OEMs are entering the intelligentization phase with the smart driving penetration increasing. This proprietary tracker analyzes China's smart driving chip industry using raw data released by Gasgoo. The dataset is based on vehicles sold to end users, as indicated by auto insurance registrations. It is important to note that there is typically a 3–5 month lead time between when smart driving SoC vendors sell chips to OEMs and Tier-1 suppliers and when the final vehicles are sold. Despite this lag, these figures provide a reliable indicator of industry trends and company performance over the medium to long term. This tracker serves as a valuable resource to help address key industry debates, including: 1) How rapidly is smart driving penetration increasing? 2) What does the competitive landscape look like? 3) Are major OEMs making changes in vendor selection or pursuing in-house development? We plan to publish this tracker quarterly, so please stay tuned for our updates.

OVERALL SMART DRIVING PENETRATION RATE

In 2Q26, the overall NOA (L2+ & L2++) penetration rebounded to 42.6%, up sharply from 32% in 1Q26, as the accelerating deployment of Urban NOA drove a broad-based improvement in smart driving adoption. Despite total passenger vehicle sales remaining 22% below the level recorded in 2Q25, vehicle sales equipped with NOA functionality increased 17.6% YoY.

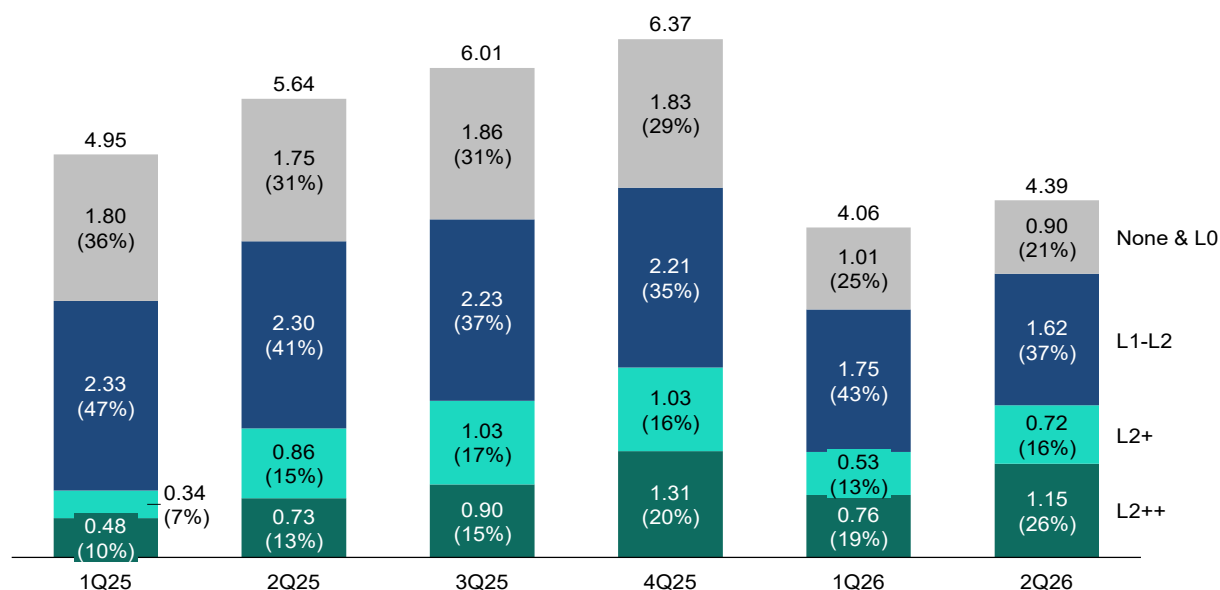
In 2Q26:

- **L1-L2 penetration declined 4ppt QoQ to 37%**, as consumers increasingly shifted toward vehicles equipped with more advanced smart driving capabilities.
- **L2+ penetration recovered 3ppt QoQ to 16%**, supported by continued rollout of cost-effective Highway NOA solutions by BYD, Geely and other domestic OEMs.
- **L2++ penetration surged 7ppt QoQ to 26%**, reaching a new record high and becoming the largest contributor to overall NOA penetration growth.

Despite the challenging PV market environment, the rebound in 2Q26 reinforces our long-term view that China's smart driving market is entering the next phase of adoption, with Urban NOA increasingly becoming a mainstream feature rather than a premium differentiator. As the industry moves beyond basic highway-assisted driving, the upgrading trend from L2+ to L2++ should continue to drive both semiconductor content growth and higher computing power requirements.

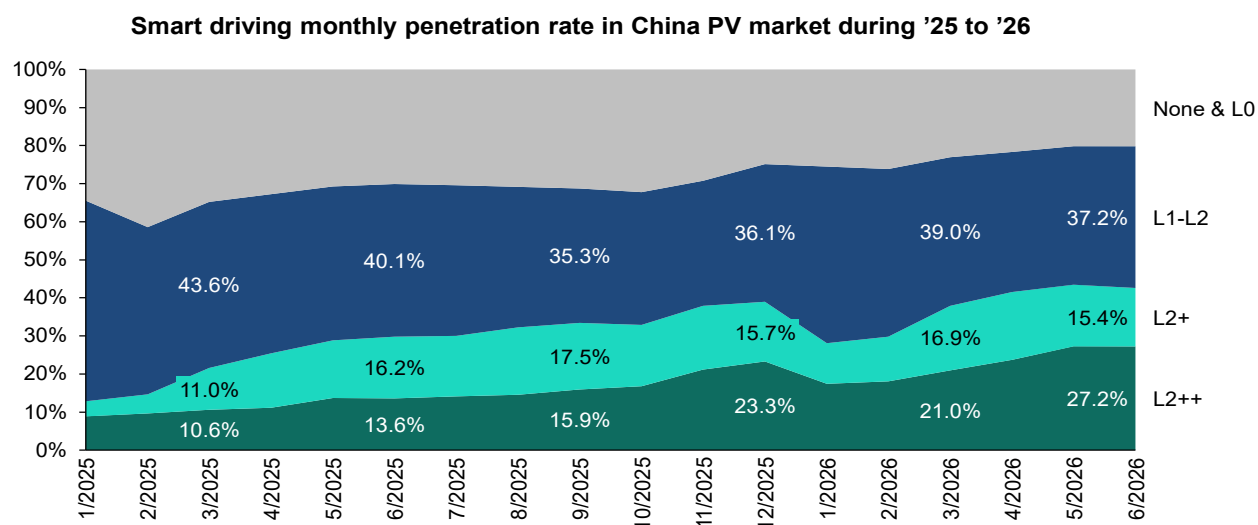
EXHIBIT 1: While China PV sales remained weak in 2Q26, record-high NOA adoption fueled 17.6% YoY sales unit growth of smart driving vehicle with L2+ & L2++ functions

Quarterly sales volume of PV in China market during '25 to '26 YTD (Mn units)



Source: Gasgoo, Bernstein analysis

EXHIBIT 2: NOA penetration rebounded rapidly following the 1Q26 dip, with L2++ penetration reaching a new record high



Source: Gasgoo, Bernstein analysis

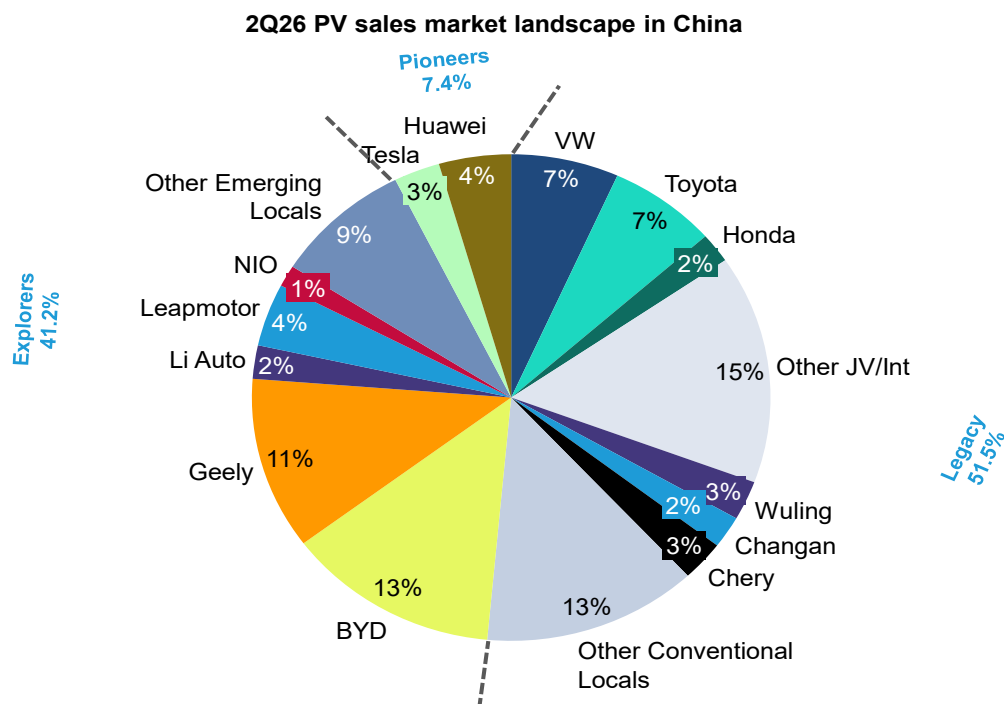
OEM MIX BY VEHICLE SALES VOLUME AND THEIR PENETRATION RATE

In 2Q26, Explorers regained market share following the seasonal weakness in 1Q26 and resumed their role as the primary driver of NOA adoption in China. Explorers PV sales outpaced the broader market, lifting their market share to 41% in 2Q26. More importantly, L2++ penetration within the Explorers rebounded sharply to 44% in June from 33.5% in March, making Explorers the single largest contributor to the recovery in overall L2++ penetration during the quarter.

- **Legacy** (JV/Int and Conventional Locals) continue to lag in advanced smart driving adoption. While NOA penetration improved to 16% in June, nearly 85% of vehicle sales still remain in the L1-L2 or without ADAS function. This suggests substantial long-term penetration headroom for both NOA features and related semiconductor content.
- **Explorers** (BYD, Geely, and Emerging Locals) remain the industry's key growth engine recently. The rapid rebound in Explorer L2++ penetration in 2Q26 reinforces our view that the segment will contribute the majority of incremental L2++ demand over the next 1-2 years. And it will also be the core battleground between OEM in-house solutions and third-party chip vendors.
- **Pioneers** (Tesla and Huawei) continue to lead the industry, with L2++ penetration remaining above 80%. However, given their already high penetration rates and relatively small share of the overall PV market, their ability to drive future industry-wide penetration expansion is increasingly limited.

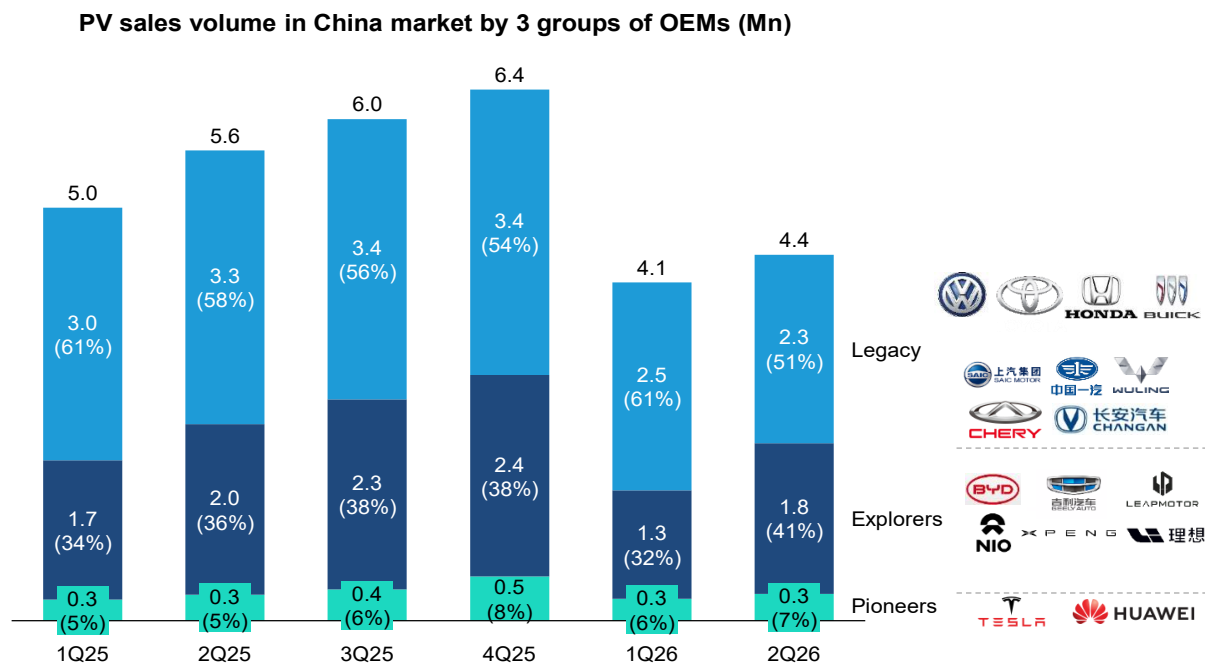
Compared with the 80-90%+ L2++ penetration already achieved by Pioneers, both Explorers and Legacy OEMs still have substantial headroom for penetration expansion. We expect Explorers to drive most of the incremental L2++ demand over the next one to two years, while the longer-term industry TAM expansion will be increasingly driven by penetration gains within Legacy OEMs. **Importantly, Legacy represents the most addressable customer segment for 3rd-party smart driving SoC vendors, given its limited in-house chip development capabilities and continued reliance on outsourced solutions.**

EXHIBIT 3: Because of difference in penetration rate and outsourcing preference, OEMs can be categorized into 3 groups



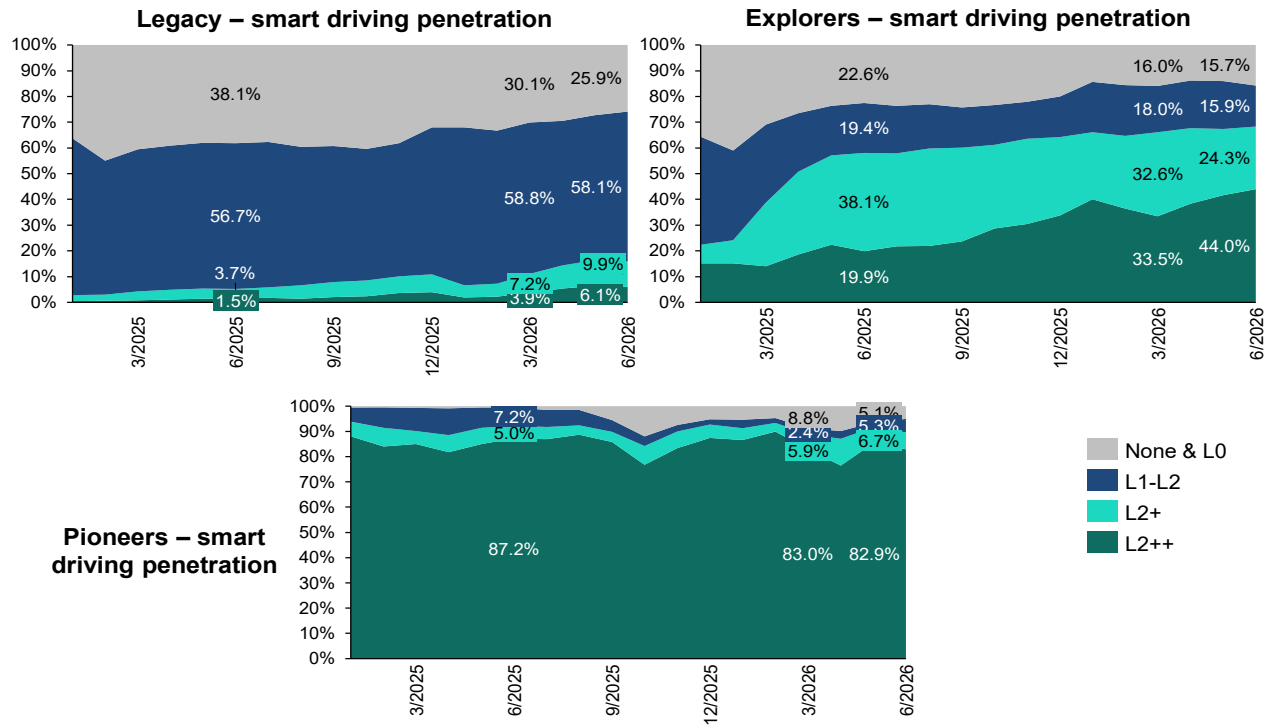
Source: Gasgoo, Bernstein analysis

EXHIBIT 4: Explorers sales rebounded more strongly in 2Q26, at the expense of the Legacy segment. As the key battleground between in-house and 3rd-party chip solutions, Explorers segment remains critical to future market landscape.



Source: Gasgoo, Bernstein analysis

EXHIBIT 5: L2++ penetration in Explorers reaccelerated in 2026, driving the overall recovery in L2++ adoption. With L2++ penetration already exceeding 80% in Pioneers, substantial headroom remains in the Legacy and Explorer segments, creating significant TAM expansion opportunities in the long run.



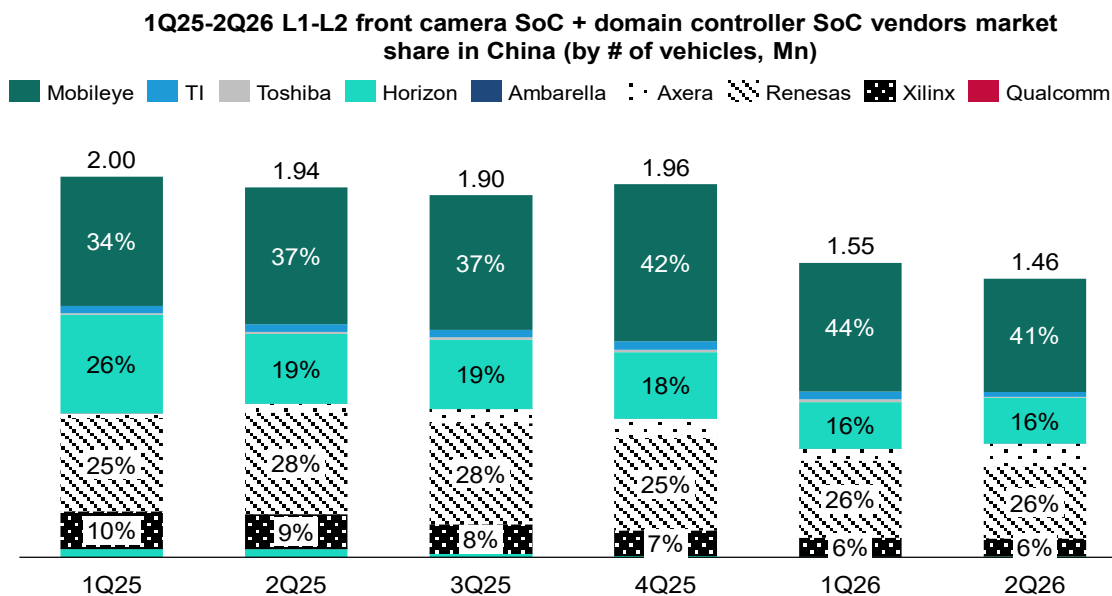
Source: Gasgoo, Bernstein analysis

VENDORS MARKET LANDSCAPE BY SMART DRIVING LEVEL

The competitive landscape for smart driving chips suppliers within **L1-L2 remains relatively stable**, with three major players commanding 15–40% market share each. However, in 2Q26:

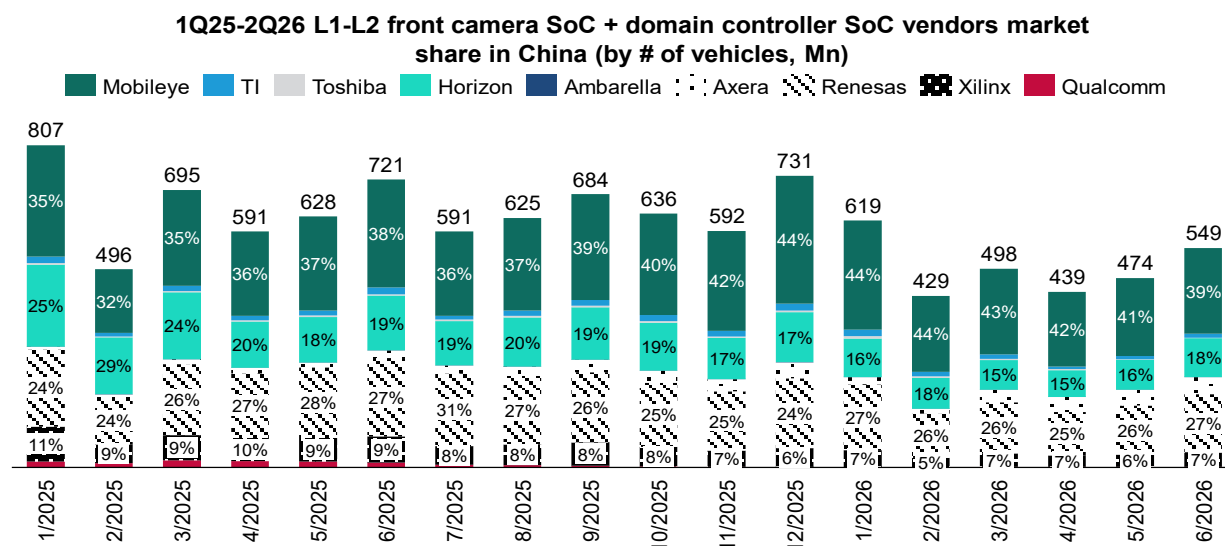
- **Horizon** stabilized its L1-L2 market share in 2Q26 as the newly launched J6B began ramping up, after several consecutive quarters of market share declines. We expect J6B to help Horizon regain share in the L1-L2 segment going forward.
- **Mobileye** retained its market-leading position in L1-L2 with a 41% market share, followed by **Renesas** at 26%.

EXHIBIT 6: **Mobileye regains the market-leading position and market share in L1-L2 segment during recent quarters, though this segment is shrinking**



Source: Gasgoo, Bernstein analysis

EXHIBIT 7: Following consecutive quarters of market share erosion, Horizon's L1-L2 market share stabilized in 2Q26, driven by the initial ramp-up of the J6B.

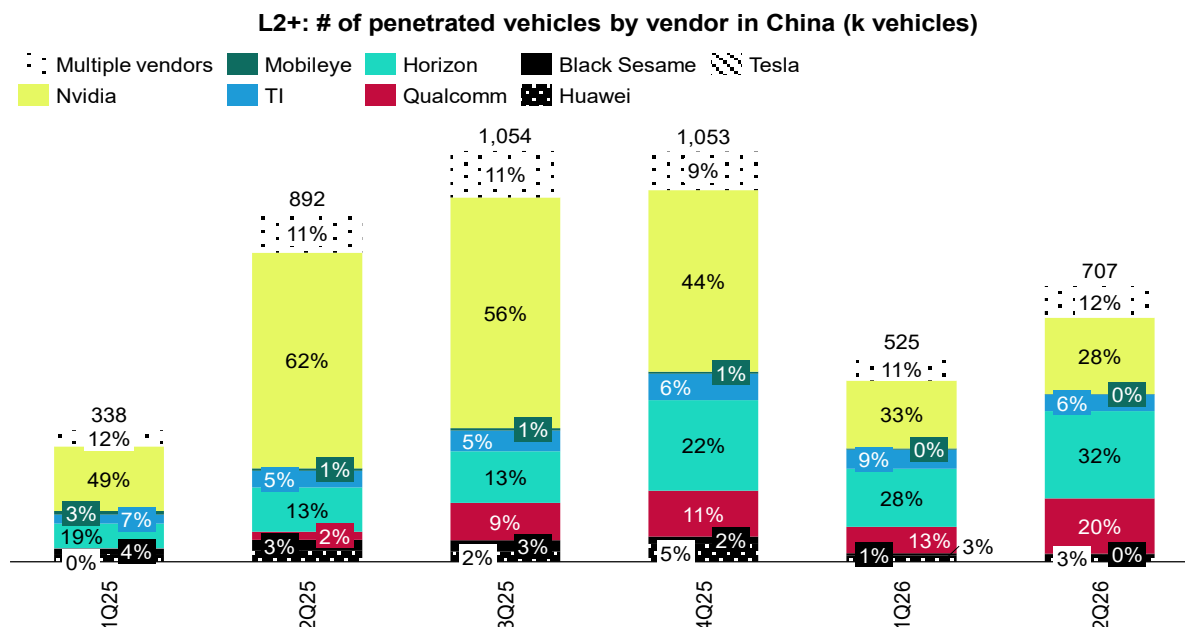


Source: Gasgoo, Bernstein analysis

Amid the recovery in the L2+ segment, it is a milestone that Horizon surpassed NVIDIA for the first time in 2Q26 to become the market leader by vehicle installations. Qualcomm also expanded its market presence, while NVIDIA continued to lose share amid intensifying competition from both emerging third-party vendors and OEM-developed solutions.

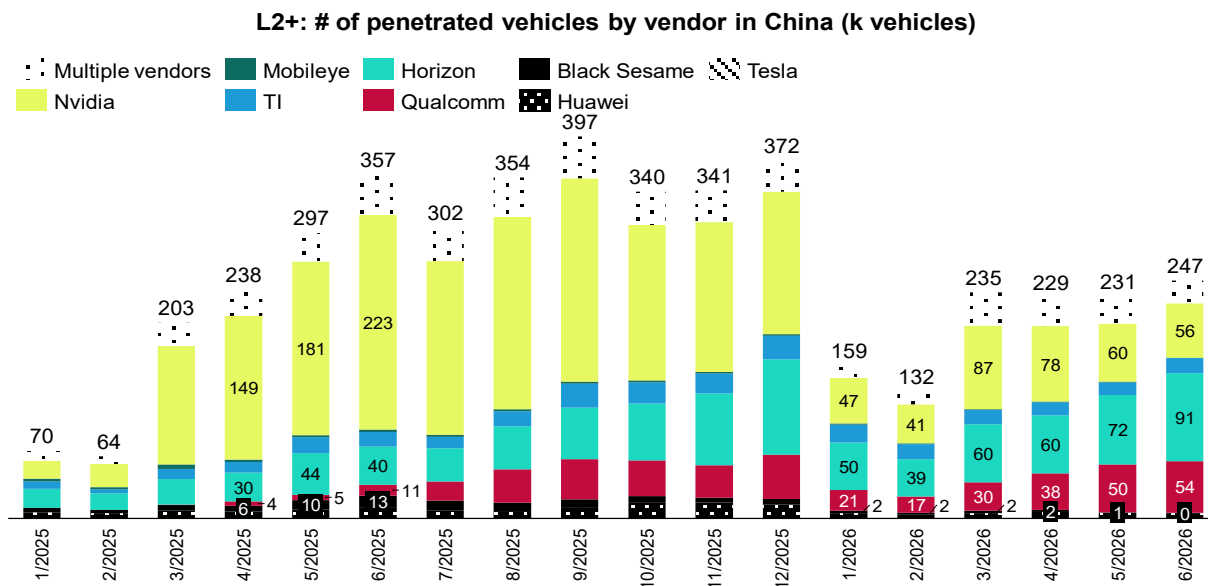
Horizon's market share reached 32% in 2Q26, exceeding NVIDIA's 28%, while Qualcomm expanded to 20%. We believe Horizon benefited from continued share gains at BYD and Geely as well as the rapid ramp-up of its J6M platform. Qualcomm's growth was primarily driven by strong deployment momentum at Leapmotor.

EXHIBIT 8: In the L2+ segment, Horizon surpassed NVIDIA for the first time in market share based on the # of equipped vehicles, during 2Q26



Source: Gasgoo, Bernstein analysis

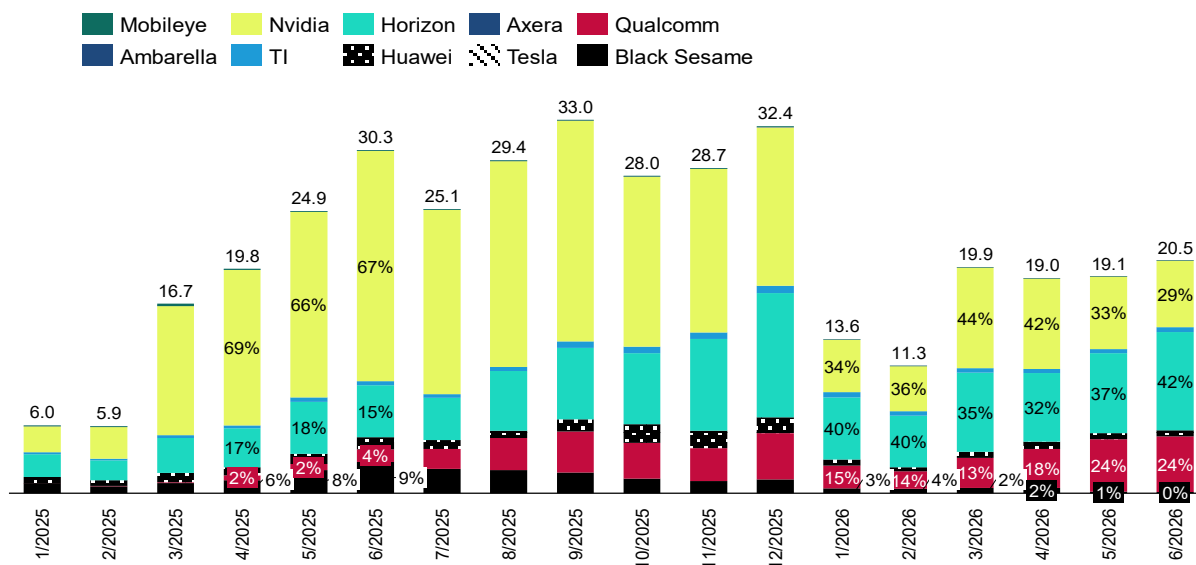
EXHIBIT 9: As the L2+ segment recovered in 2Q26, Horizon emerged as the market leader while Qualcomm also gained momentum, with both vendors taking share from NVIDIA



Source: Gasgoo, Bernstein analysis

EXHIBIT 10: In L2+, market landscape measured by TOPS shows a trend similar to that by # of vehicles

L2+: total TOPS installed in sold vehicles by vendor in China (Mn TOPS)



Source: Gasgoo, Bernstein analysis

Within the L2++ segment, Nvidia continues to hold a leading position. However, in recent quarters, emerging 3rd-party vendors, such as Horizon and Qualcomm, and OEMs' in-house solutions have seen rising adoption, intensifying the competition.

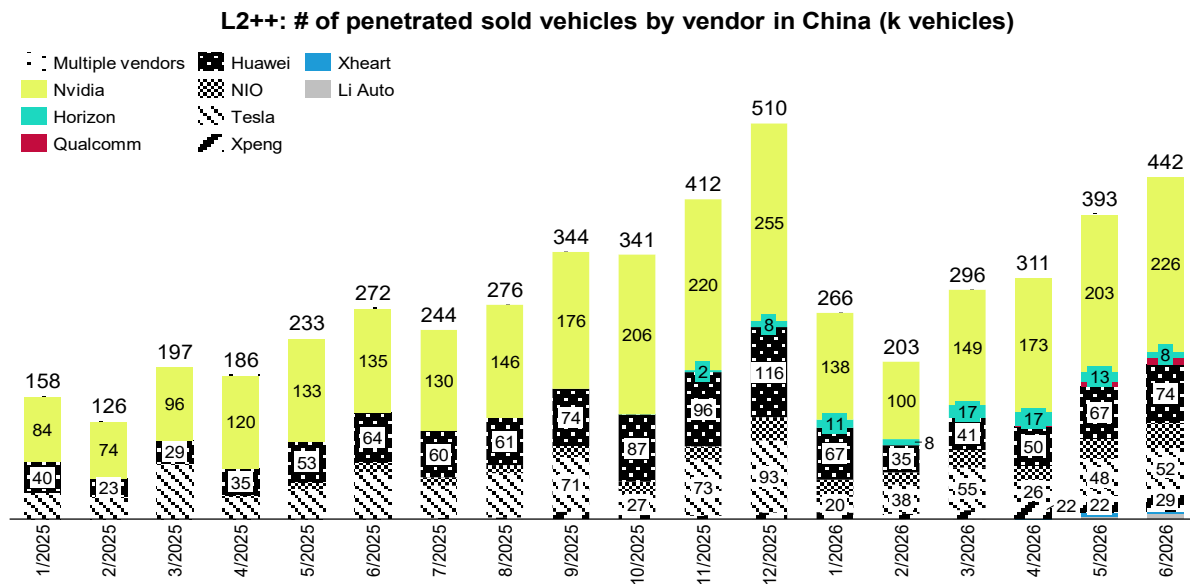
Compared with the L2+ segment, competition in L2++ remains at a much earlier stage. NVIDIA continues to lead the market with a 52% share by vehicle installations in 2Q26, but its dominance is gradually being challenged by both third-party vendors and OEM-developed solutions. Horizon's share declined sequentially to 3%, reflecting slower-than-expected design win progress, while Qualcomm formally entered the segment through the deployment of the SA8797P platform.

Since 2Q26, Li Auto has begun deploying its in-house M100 SoC (1,280 TOPS). Through single- and dual-M100 configurations, the platform supports centralized computing across smart driving and cockpit domains in the latest L8 and L9 models.

Meanwhile, Xheart, Momenta's chip-design affiliate, has started deploying its X7 chip (272 TOPS), with Volkswagen's ID. ERA becoming the first production vehicle to adopt the solution. While Xheart benefits from Momenta's strong software capabilities, we believe its late entry into the L2++ hardware market substantially limits its addressable market opportunity, given the increasingly entrenched positions of NVIDIA, Horizon, Qualcomm, and leading OEM in-house solutions. More details can be found in our [published competitor analysis report](#).

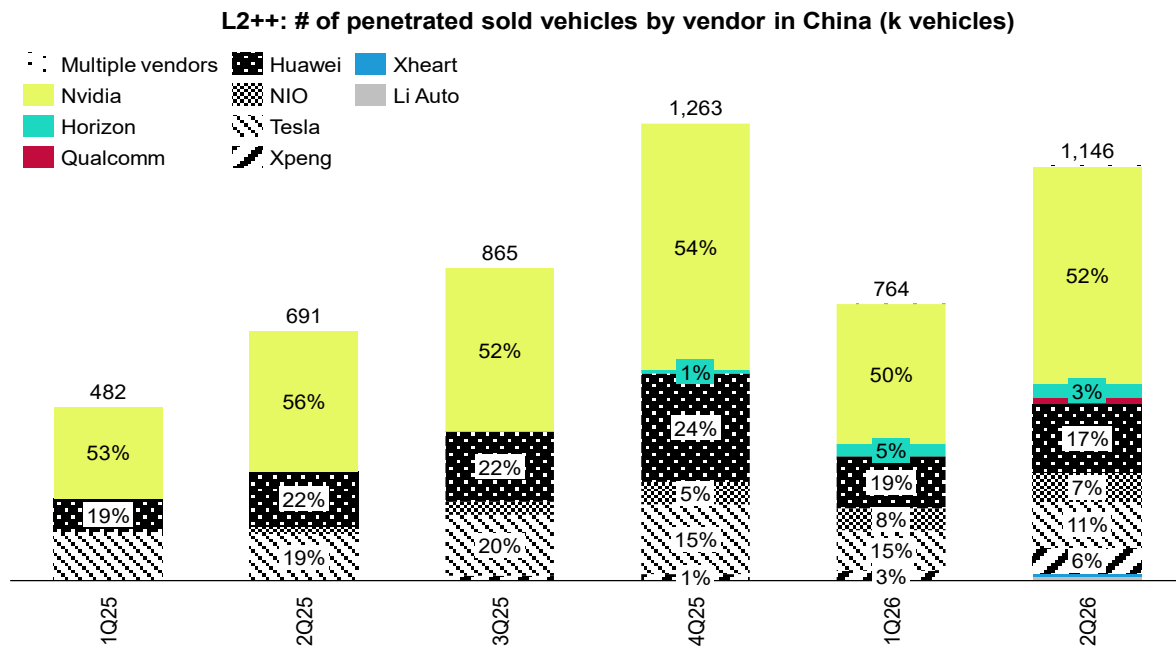
We expect the next phase of competition in L2++ to be driven primarily by BYD's supplier allocation strategy, as the company rapidly expands Urban NOA penetration. The ability of Horizon to secure meaningful wallet share at BYD will likely determine whether NVIDIA can defend its leading position over the next two years.

EXHIBIT 11: Within the L2++ segment, NV remains the market leader. However, rising adoption of emerging 3rd-party vendors and Explorers' in-house solutions is increasing competitive intensity



Source: Gasgoo, Bernstein analysis

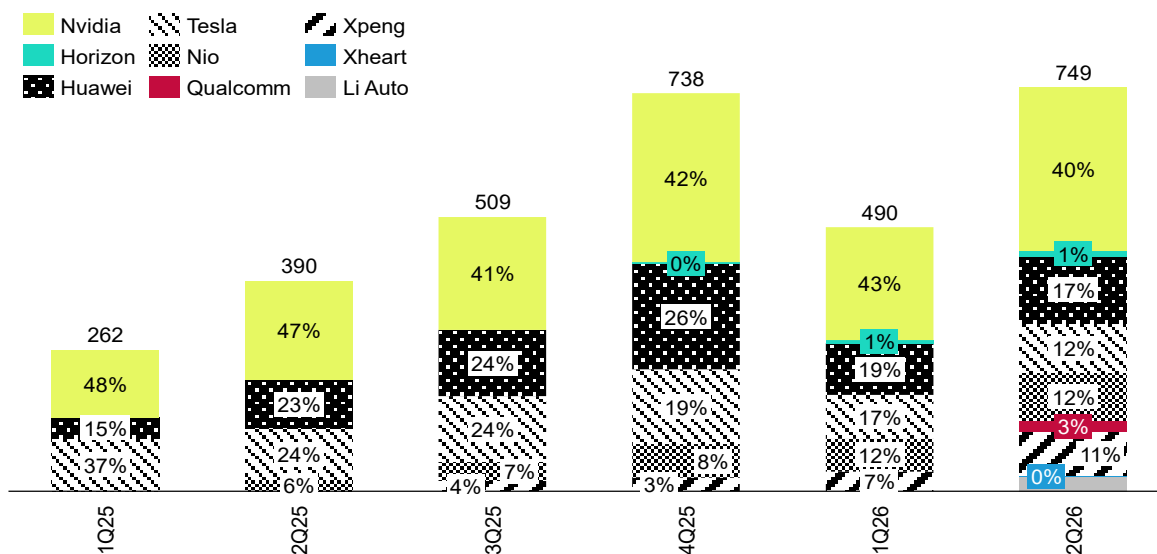
EXHIBIT 12: Amid intensifying competition, Horizon's progress in securing L2++ design wins has been relatively slow, resulting in a sequential market share drop in 2Q26



Source: Gasgoo, Bernstein analysis

EXHIBIT 13: **Horizon's L2++ share is smaller on a TOPS basis than on a vehicle basis, as OEMs with in-house solutions generally deploy higher-TOPS chips, suggesting lower optimization efficiency. Qualcomm, meanwhile, commands a larger TOPS share relative to vehicle share due to its dual-SoC central computing architecture.**

L2++: total TOPS installed in sold vehicles by vendor in China (Mn TOPS)



Source: Gasgoo, Bernstein analysis

MAJOR VENDORS CUSTOMERS & PRODUCTS DEEP DIVE

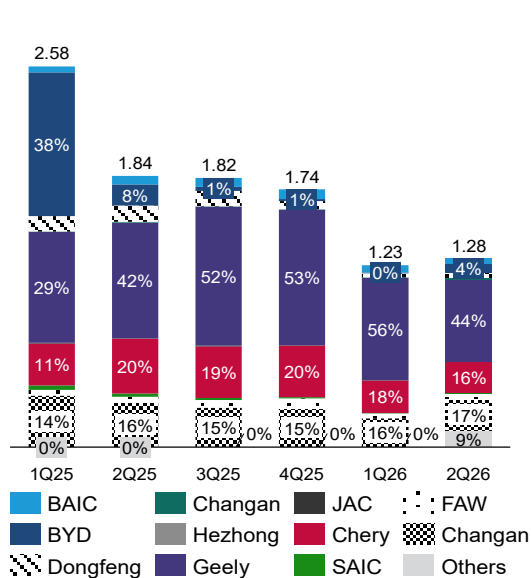
HORIZON:

- **Geely has surpassed BYD as Horizon's largest customer in L1-L2 front camera chips and domain controller SoCs**, contributing 44% of Horizon's total TOPS in 2Q26. This shift is due to BYD's focus shifting to L2+. We expect that the newly-released J6B will help Horizon gain new design wins in L1-L2.
- **In the L2+ segment, Horizon has begun replacing NVIDIA in selected BYD models and targets achieving 90%+ wallet share in BYD's GodEye C platform by 2H26-2027.** Given BYD's dominant 35-40% share of the L2+ market, such adoption guidance could lift Horizon's L2+ market share from current 32% to 55-60%, assuming its wallet share across other OEMs remains broadly stable.
- **In the L2++ segment, adoption by BYD in its L2++ offerings represents a meaningful near-term upside for Horizon.** A single J6M or dual-J6M configuration is increasingly being positioned as an alternative to J6P, offering a more compelling cost-performance proposition in the current tough market environment. This transition reduces Horizon's content value per vehicle. However, the disclosed collaboration with BYD in its GodEye B models has potential to bring significant upside. **Given BYD's 16% share of the L2++ market, securing even one-third wallet share could nearly triple Horizon's L2+ market share from ~3% currently.** The recent price correction has not fully priced in this substantial upside.

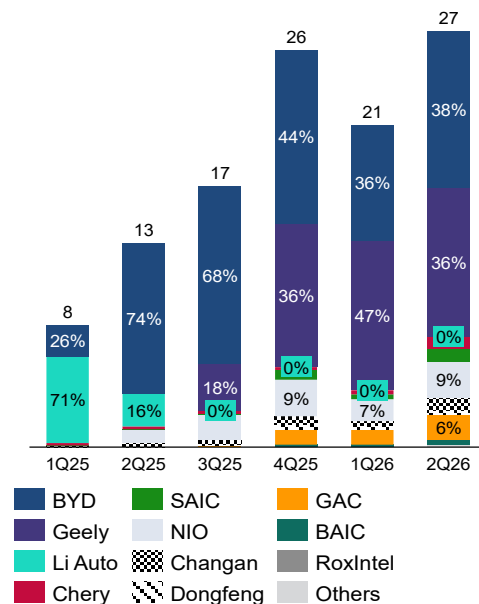
Going forward, a key factor to monitor will be how BYD allocates its L2++ vehicle models among Horizon, NVIDIA, and its in-house chip solutions.

EXHIBIT 14: In traditional L1-L2 and L2+ segments, BYD and Geely continue to contribute the majority of revenue to Horizon. But Horizon is diversifying its client pool by acquiring new OEM clients every quarter

Horizon's customer mix by TOPS in L1-L2 domain controller SoC + front camera chips (Mn TOPS)

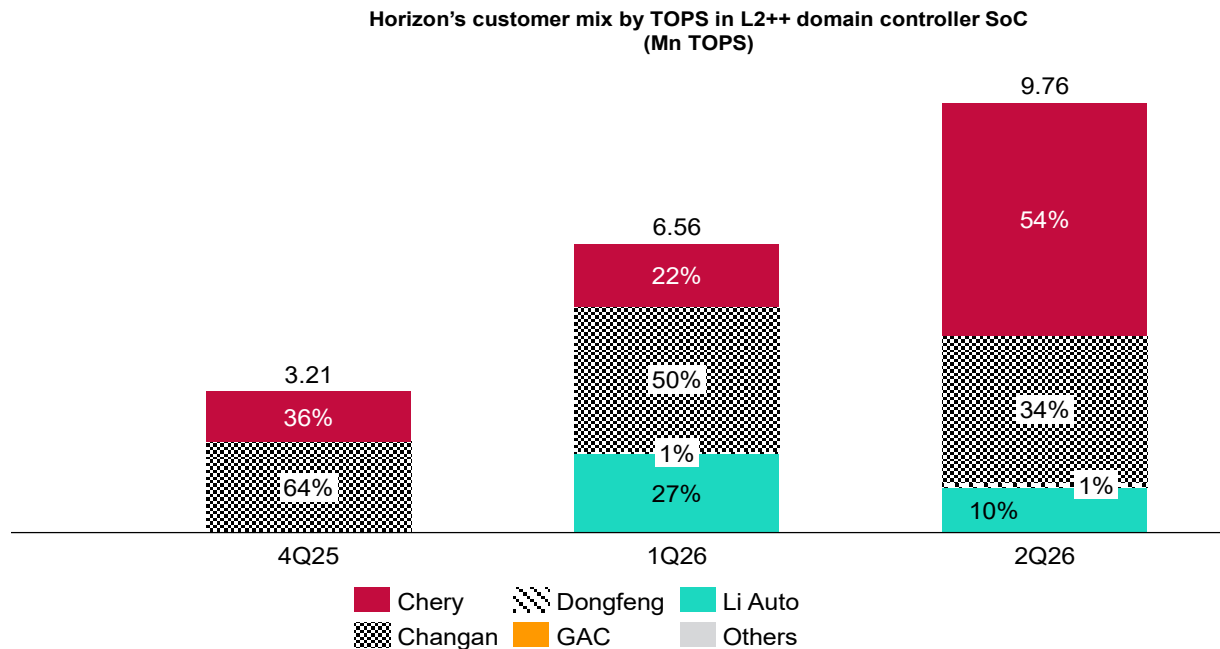


Horizon's customer mix by TOPS in L2+ domain controller SoC (Mn TOPS)



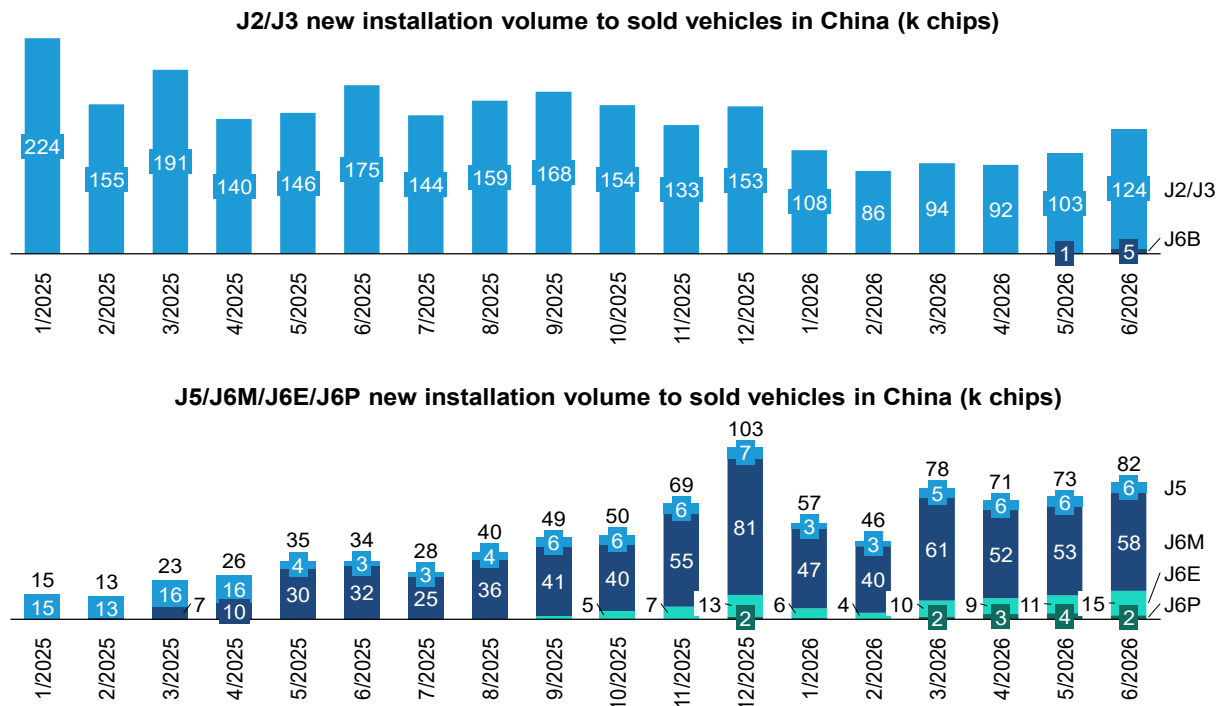
Source: Gasgoo, Bernstein analysis

EXHIBIT 15: In the emerging L2++ segment, Chery, Changan, and Li Auto are currently Horizon's core customers. But J6P rollout lags our initial expectations, with only one new Chery model (Fulwin T9L) deployed as of 2Q26



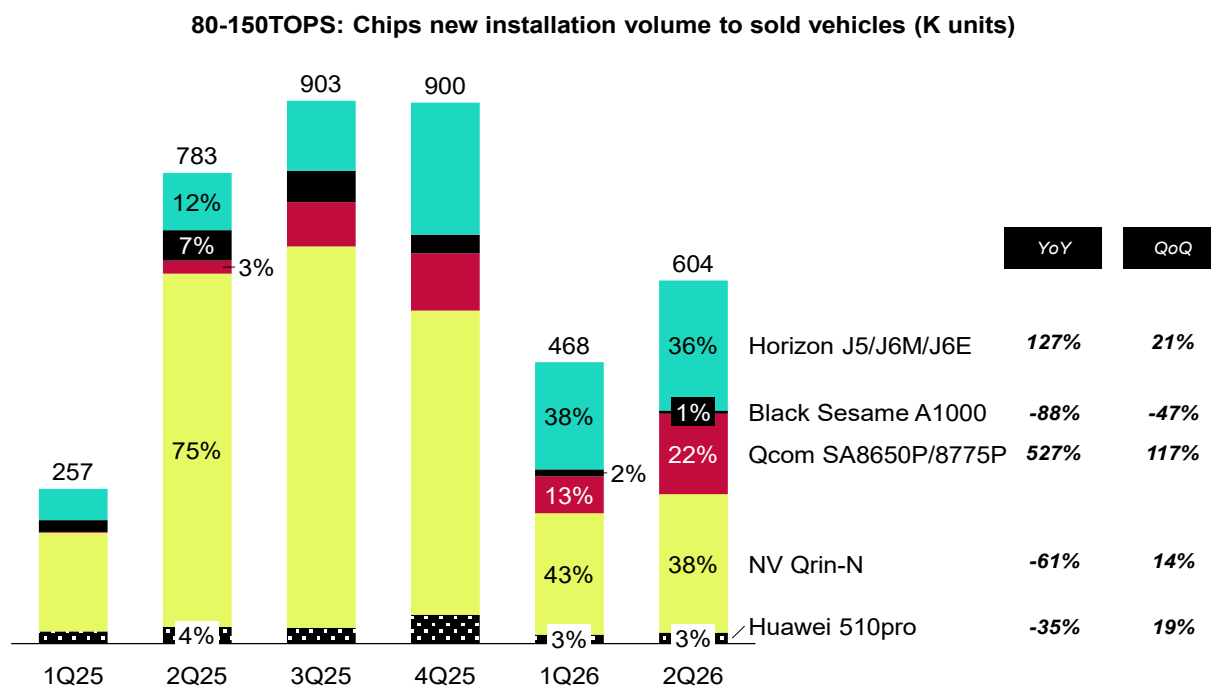
Source: Gasgoo, Bernstein analysis

EXHIBIT 16: Horizon's installation volumes rebounded sequentially across both the below-20 TOPS and above-100 TOPS categories in 2Q26, after a steep contraction in 1Q



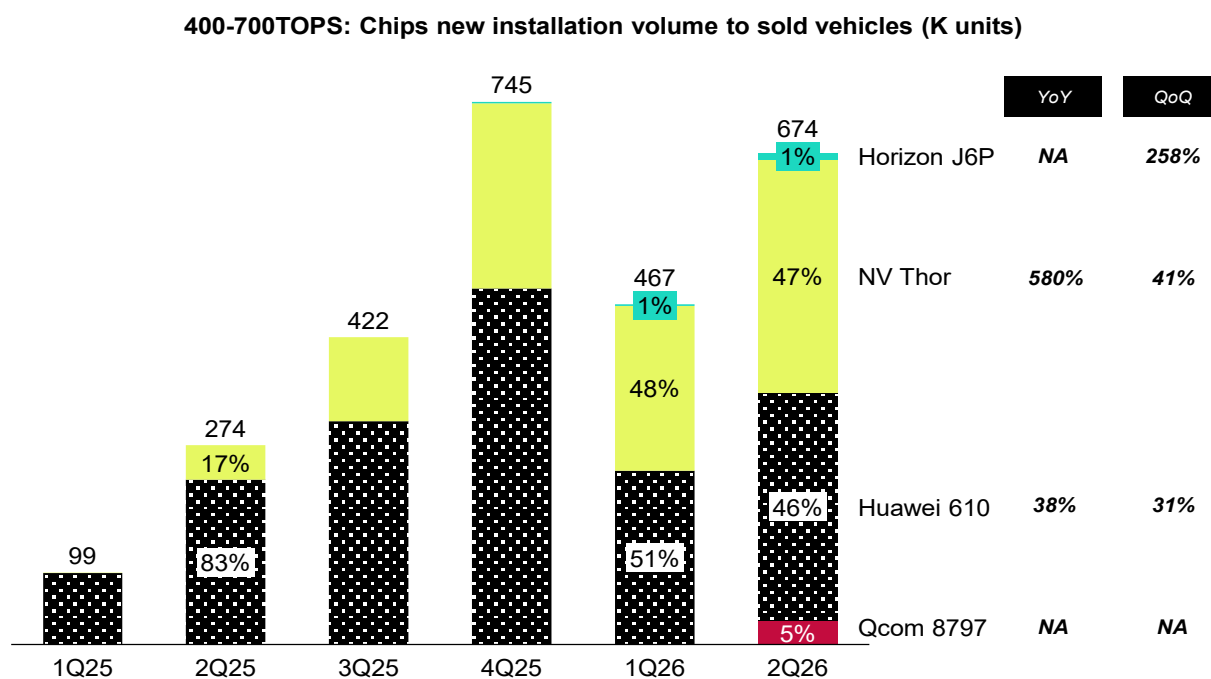
Source: Gasgoo, Bernstein analysis

EXHIBIT 17: **Within the 80-150 TOPS segment, Horizon, along with Qualcomm, led the market recovery with positive YoY growth, while most competing products remained below their 2Q25 installation levels**



Source: Gasgoo, Bernstein analysis

EXHIBIT 18: **Among outsourced 400-700 TOPS solutions, Horizon J6P and Qualcomm SA8797P posted the strongest installation growth, benefiting from a low base**



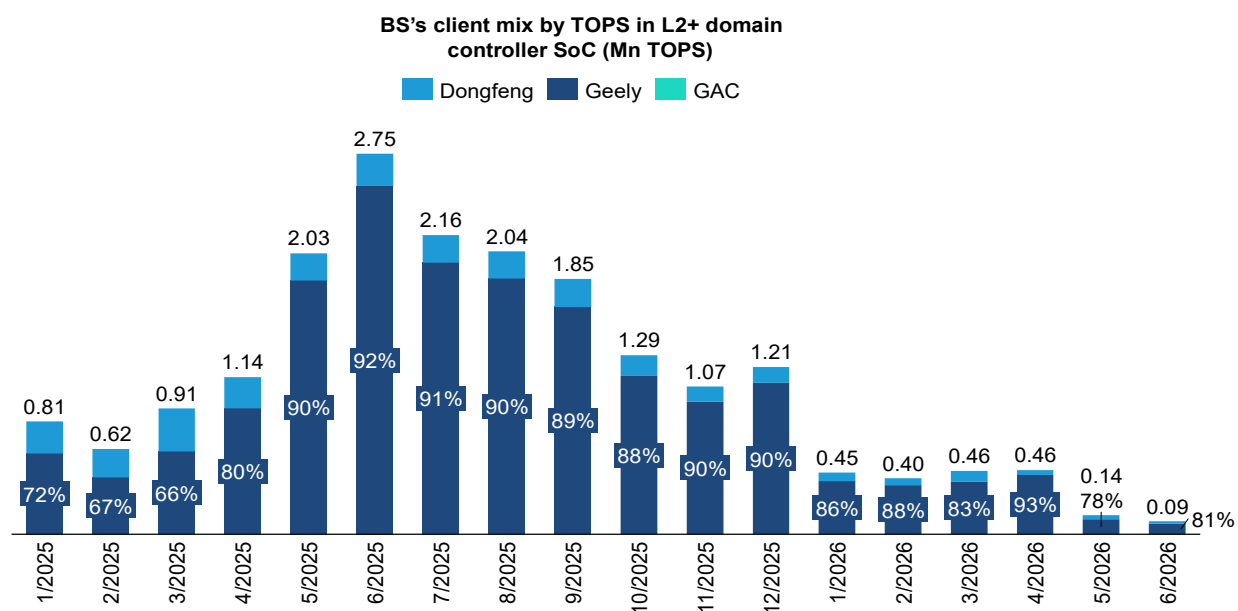
Source: Gasgoo, Bernstein analysis

BLACK SESAME:

- **Black Sesame's expansion remains largely tied to Geely, which contributes ~80-90% of its sales volume.**
- **The A1000 (96 TOPS) has recorded sequential declines for four consecutive quarters**, primarily due to the weak performance of Geely models equipped with the A1000. Furthermore, Geely is allocating a greater share of its business to Horizon.

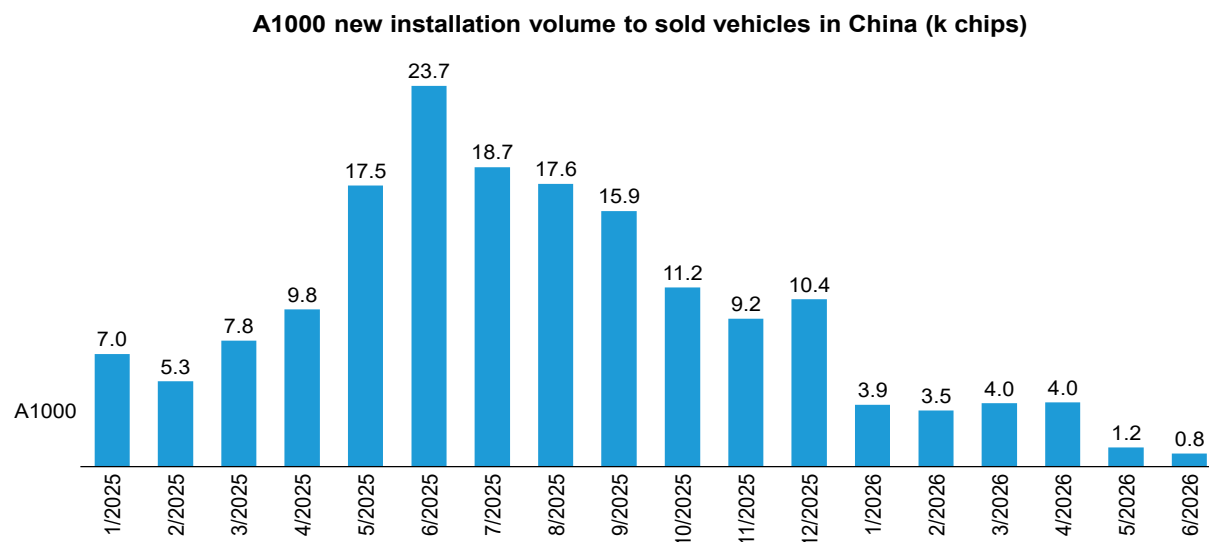
Post Zhuhai EEasyTech acquisition, Black Sesame is likely to gain share in the L1–L2 entry-level market. C1200 design win progress is a key catalyst to watch.

EXHIBIT 19: Without acquiring new clients till 2Q26, Black Sesame has achieved installation in only 3 OEMs (Dongfeng, Geely, and GAC), with a heavy reliance on Geely



Source: Gasgoo, Bernstein analysis

EXHIBIT 20: A1000 continues to experience a sequential decline, due to poor sales performance of Geely's models embedded with A1000 and Geely's wallet share shift to Horizon



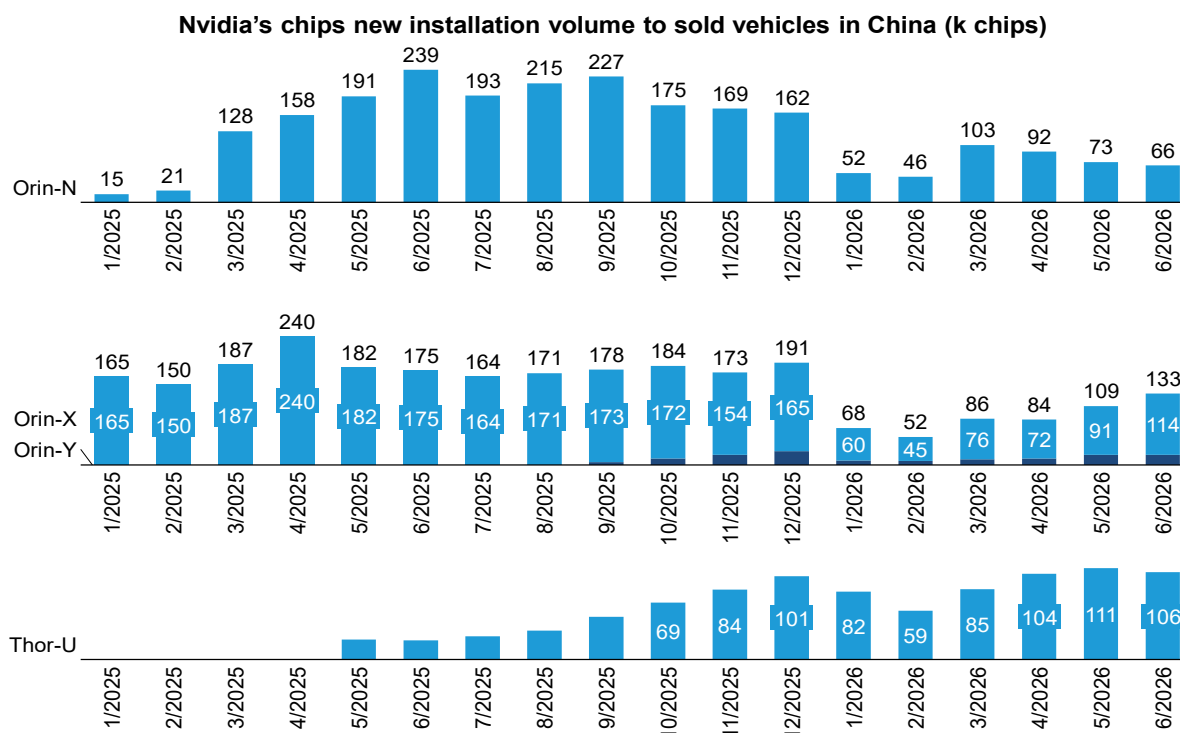
Source: Gasgoo, Bernstein analysis

NVIDIA:

- In 2Q26, new installations of Orin-N (84 TOPS) remained well below year-ago levels, as Horizon's J6M and J6E continued to gain meaningful share from NVIDIA in the L2+ segment.
- Monthly installation volumes of Orin-X and Orin-Y rebounded sequentially, although they remained below the peak levels achieved in 2025.
- The ramp-up of Thor-U (700 TOPS) installations has also moderated.

Bernstein expects a prolonged share battle between NVIDIA's Orin-N and Horizon's J6M/J6E in L2+. In L2++, NVIDIA is likely to see its market leadership gradually erode as emerging third-party vendors and OEM-developed solutions gain traction.

EXHIBIT 21: All NVIDIA product lines recorded a sequential rebound in 2Q26, reflecting the broader market recovery. However, installation volumes remained below the peak levels reached in 2H25, as competition from alternative third-party vendors and OEMs' in-house solutions continued to intensify



Source: Gasgoo, Bernstein analysis

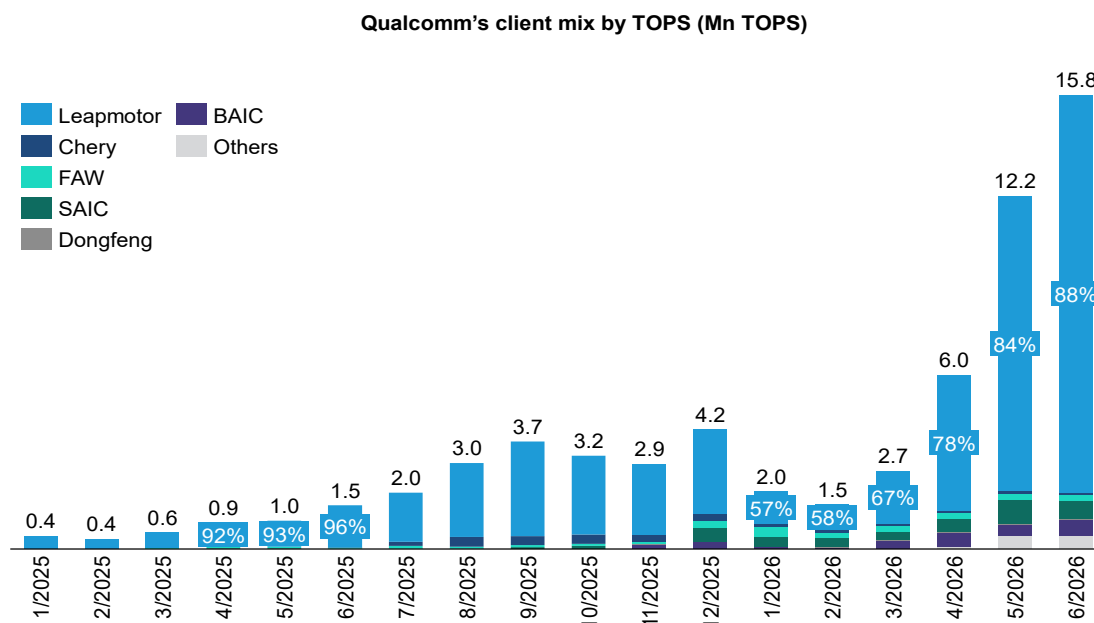
QUALCOMM:

Qualcomm substantially increased its L2+ market share from 13% to 20% QoQ in 2Q26. The company added one new OEM customer during the quarter, bringing its total OEM customer base to seven, while Leapmotor continues to account for the majority of its deployment volume.

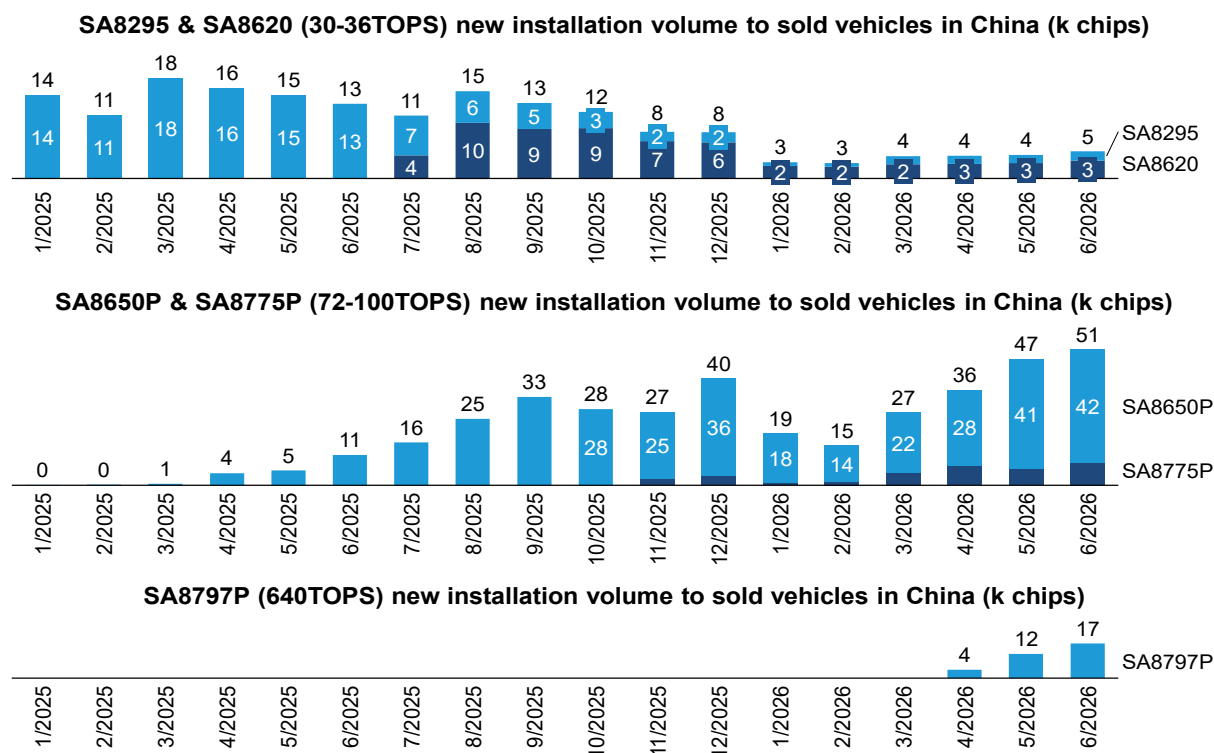
The launch of SA8797P (600+ TOPS) also officially established Qualcomm's presence in the L2++ segment.

Qualcomm's strong momentum has been driven primarily by its partnership with Leapmotor. Leapmotor's value-for-money positioning aligns well with Qualcomm's auto offerings.

- To optimize R&D and procurement costs, Leapmotor standardizes its E/E architecture across vehicle platforms, enabling Qualcomm to secure design wins across the company's entire model lineup and significantly accelerate deployment ramp-up. In contrast, Horizon has typically secured design wins on a model-by-model basis at OEMs such as Chery and Deepal, which follow a multi-vendor sourcing strategy. As a result, Horizon's deployment pace has been slower than Qualcomm's, despite its strong competitive positioning. That said, sustaining Qualcomm's current growth trajectory will likely depend on its ability to diversify beyond its heavy reliance on Leapmotor.
- From a technology perspective, single- or dual-SA8797P configurations support a centralized computing architecture that integrates smart driving, cockpit, and vehicle-body control functions. This cross-domain architecture enables meaningful BOM cost savings and is becoming increasingly attractive amid rising memory costs and intense pricing competition in the automotive industry. Qualcomm's success highlights the growing market potential of integrated smart driving-and-cockpit SoCs, and reinforces our positive outlook on Horizon's upcoming Starry platform ([see our published report on Starry for further details](#)).

EXHIBIT 22: **Though diversify its customer base, Qualcomm still heavily relies on Leapmotor**

Source: Gasgoo, Bernstein analysis

EXHIBIT 23: **SA8797P (600+ TOPS) began volume vehicle deployments in 2Q26, marking Qualcomm's entry into the premier tier of smart driving chip vendors alongside NVIDIA, Huawei, and Horizon**

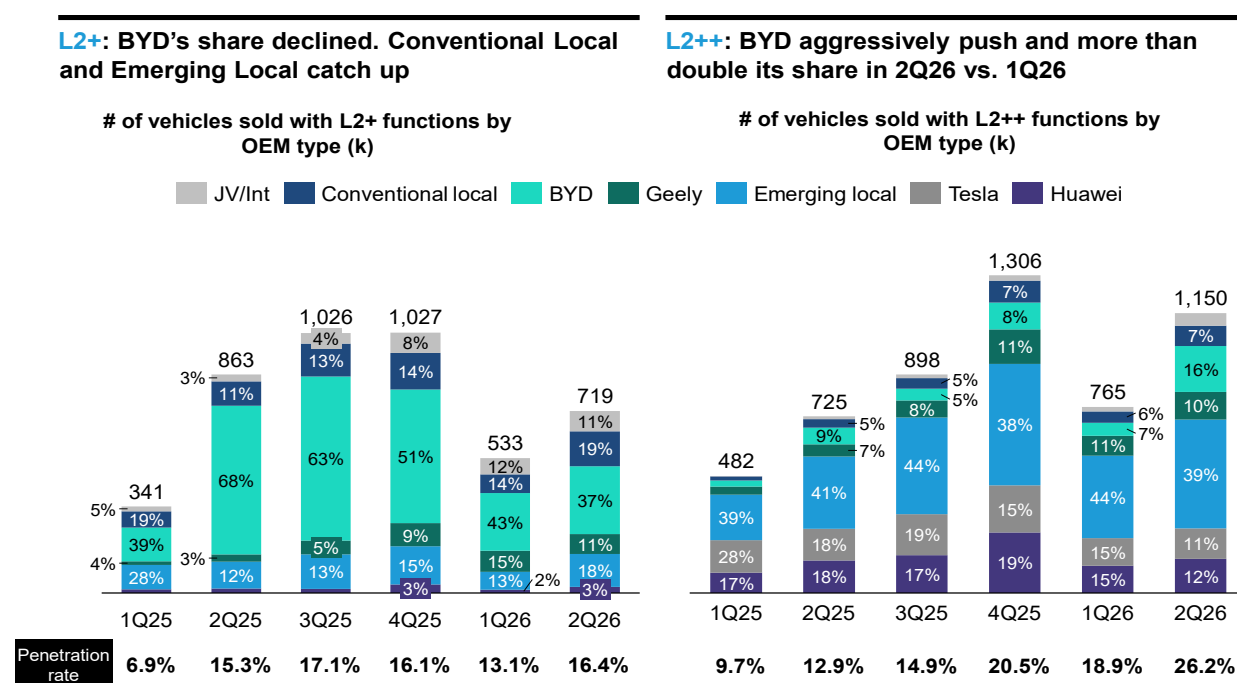
Source: Gasgoo, Bernstein analysis

OEM SHARE BY SMART DRIVING TIERS

SALES VOLUME OF VEHICLES WITH SMART DRIVING TIERS BY OEM:

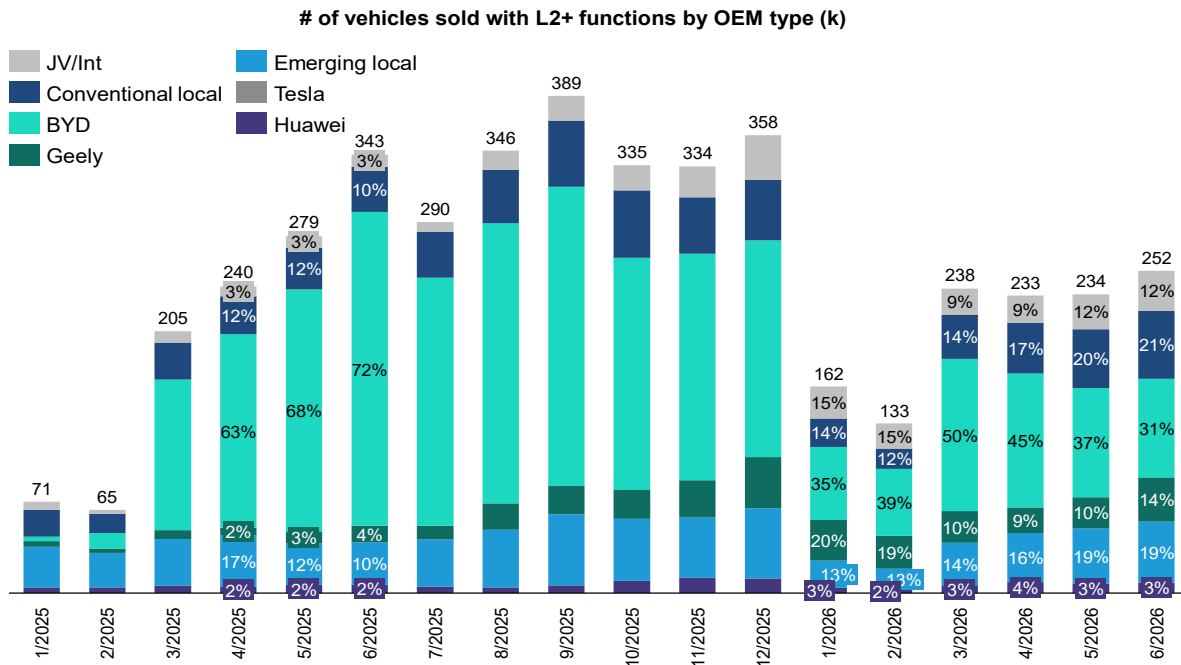
- In L2+, BYD remained the largest OEM group with approximately 37% share in 2Q26, although its dominance has gradually declined as Conventional Local and Emerging Local OEMs accelerated adoption of Highway NOA. This trend should benefit 3rd-party chip suppliers, since most Conventional Local OEMs continue to rely on outsourced smart driving SoCs.
- In L2++, BYD nearly doubled its market share versus 1Q26 and became the largest source of incremental Urban NOA demand. Emerging Local OEMs continue to lead the segment overall, while Huawei and Tesla remain important contributors. Given BYD's accelerating rollout of Urban NOA functions, progress in supplier allocation among NVIDIA, Horizon and BYD's in-house solutions will be one of the most important variables to monitor in the coming quarters.

EXHIBIT 24: In L2+, BYD's share contracted to ~40%, while Conventional Local and Emerging Local caught up. In L2+, BYD rapidly expanded its share in L2++ during 2Q26



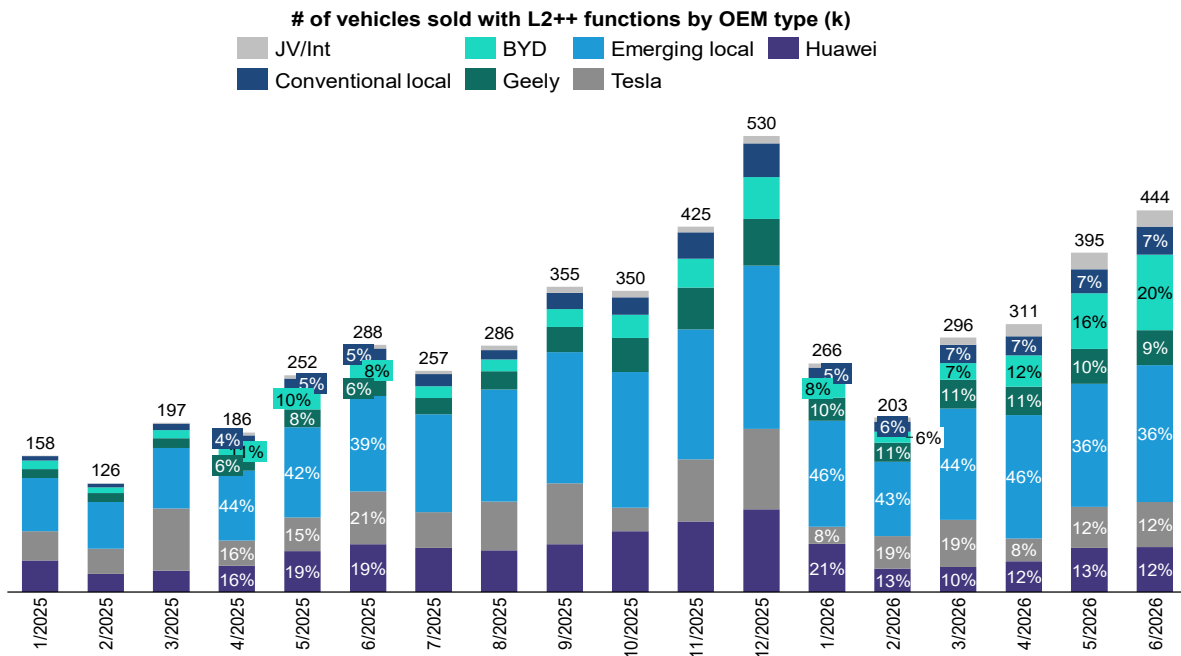
Source: Gasgoo, Bernstein analysis

EXHIBIT 25: Following BYD's trajectory, Conventional Local and Emerging Local OEMs are rapidly gaining share in the L2+ segment. As Conventional Locals are likely to remain largely reliant on outsourced smart driving SoCs, the trend should create favorable opportunities for 3rd-party vendors such as Horizon.



Source: Gasgoo, Bernstein analysis

EXHIBIT 26: BYD nearly doubled its share in L2++ segment versus 1Q26, making it the largest incremental L2++ demand in the near term. As a result, tangible progress in Horizon's collaboration with BYD on the GodEye B represents the most core near-term upside catalyst



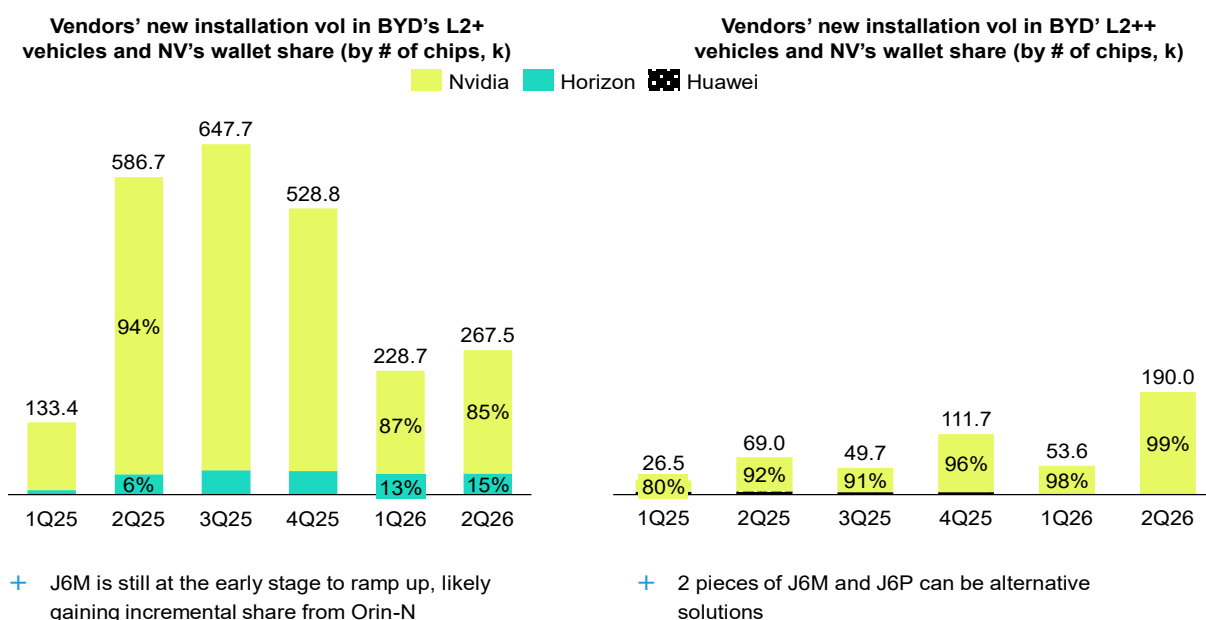
Source: Gasgoo, Bernstein analysis

SMART DRIVING CHIP VENDOR SHARE WITHIN KEY OEMS

BYD:

- Nvidia remains the primary supplier for L2+ and L2++ SoCs at BYD, while Horizon's presence in the L2+ segment is ramping up.
- Alongside Han EV and Yuan UP as the initial models utilizing the J6M, Qin L EV and Seal begin adopting Horizon J6M as a secondary source starting from 3Q25, partially replacing NVIDIA Orin-N. Given J6M's competitive advantages in both computing power and cost efficiency over Orin-N, we expect this trend of switching to continue. Horizon management guided that Horizon will obtain 90-100% wallet share in BYD L2+ models by the end of 2026 to 2027.
- In the L2++ space, BYD nearly quadrupled its SoCs deployment volume QoQ in 2Q26, with Nvidia predominantly accounting for 99% of those. Rest goes to Huawei, as Fangchengbao brand (2025 version of Bao 5 and Bao 8) use Huawei chips. Per Horizon's management, BYD will employ triple sources in L2++ SoCs, including in-house solution, NVIDIA, and Horizon. Adoption of Horizon L2++ by BYD can be the most critical catalyst to Horizon stock price in the next 12 months.

EXHIBIT 27: BYD mainly sources from NVIDIA currently. Horizon begins being adopted as the 2nd-source supplier, and its replacement to NVIDIA will accelerate in upcoming quarters, per Horizon management guidance

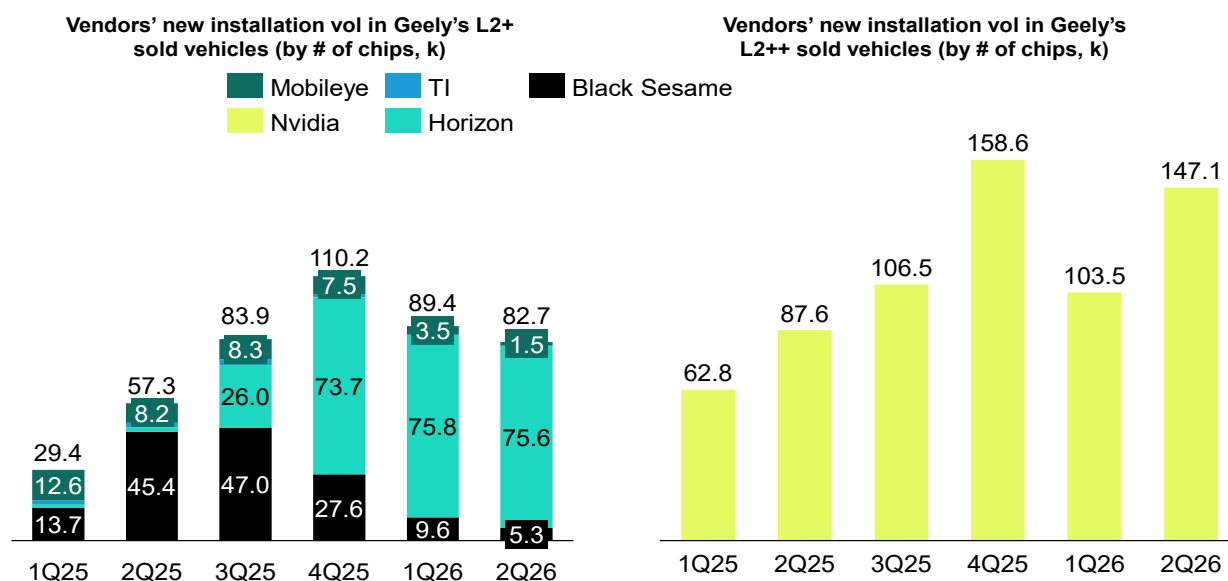


Source: Gasgoo, Bernstein analysis

Geely:

- Geely sources L2+ SoCs from a variety of suppliers but relies solely on Nvidia for L2++ SoCs.
- Black Sesame is primarily used in select Lynk and Galaxy models, many of which experience weak sales performance or are being phased out. An increasing number of Geely Galaxy models are adopting Horizon J6M as their L2+ solution. This shift in Geely's procurement strategy may limit Black Sesame's sales, as approximately 90% of its installed volume is currently contributed by Geely.
- Whether Horizon can provide L2++ solution to Geely can be another critical catalyst to watch.

EXHIBIT 28: Geely allocates more incremental volume to Horizon in L2+, resulting in the decline of Black Sesame's shipment volume. Meanwhile, Geely procures L2++ SoCs exclusively from NVIDIA



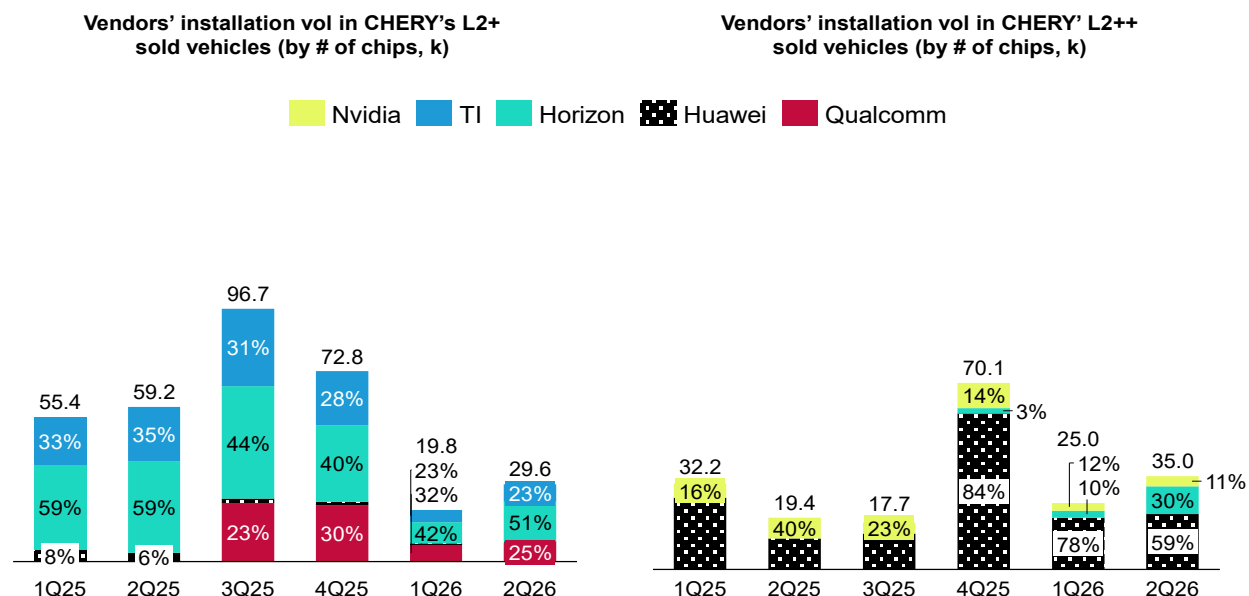
Source: Gasgoo, Bernstein analysis

Chery:

Chery adopts a three-vendor procurement strategy for smart driving chips, likely due to the group's diverse range of sub-brands and model platforms.

- In the L2+ segment, Chery primarily sources SoCs from Horizon, TI, and Qualcomm. Horizon contributed 15.1k installations in 2Q26, capturing roughly 51% of Chery's L2+ wallet share. Horizon and TI solutions are deployed across various models under the iCAR and Exeed brands. Qualcomm is ramping up its wallet share rapidly in recent quarters.
- For L2++, Chery sources from Nvidia, Horizon, and Huawei. Installations from Horizon and Huawei have outpaced those of Nvidia, suggesting growing confidence in domestic suppliers. In 2Q26, Chery has begun deploying Horizon's J6P in its newly released Fulwin T9L, the third model with J6P after the Exeed ET5 and iCAR V27. Looking ahead in upcoming quarters in 2026, approximately 10 Exeed models are guided by Horizon management to adopt Horizon's J6P for L2++, positioning Horizon for a meaningful acceleration in installation volume.

EXHIBIT 29: Chery adopts a three-vendor procurement strategy for smart driving chips, with Horizon rising to the runner up position in L2++ and likely to expand wallet share. The Fulwin T9L has become the 3rd Chery model deploying J6P since 2Q26



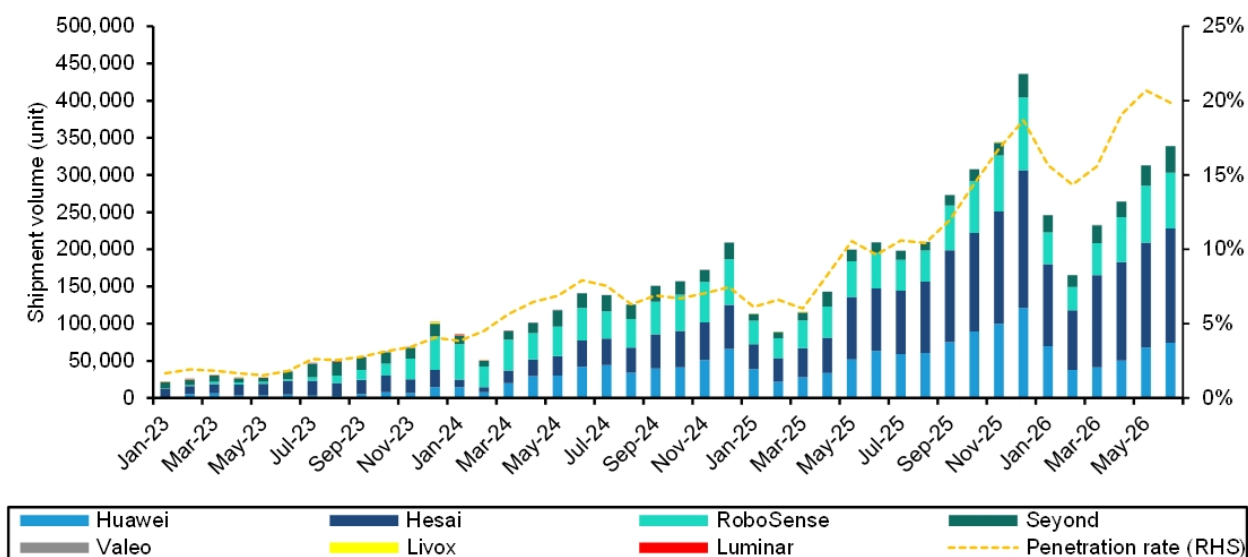
Source: Gasgoo, Bernstein analysis

LIDAR

LiDAR is seeing rapid adoption in China (Exhibit 30), with penetration rebounded to 20% in 2Q2026 (vs. ~15% in 1Q2026). Hesai continued to lead the market with 47% share in 2Q2026 (Exhibit 31), followed by RoboSense (23%), Huawei (21%), and Seyond (9%). Today, in China, more than 80% of vehicle models equipped with Urban NOA (Navigate on Autopilot) use LiDAR, far surpassing adoption rates with lower-level ADAS (Advanced Driver-Assistance Systems) features. The continued upgrade of smart driving capabilities will be a major driver of further LiDAR penetration. As the global leader in LiDAR, Hesai is well positioned to capture this structural growth trend.

EXHIBIT 30: Long-range ADAS LiDAR in China: Increasing penetration and volume share breakdown

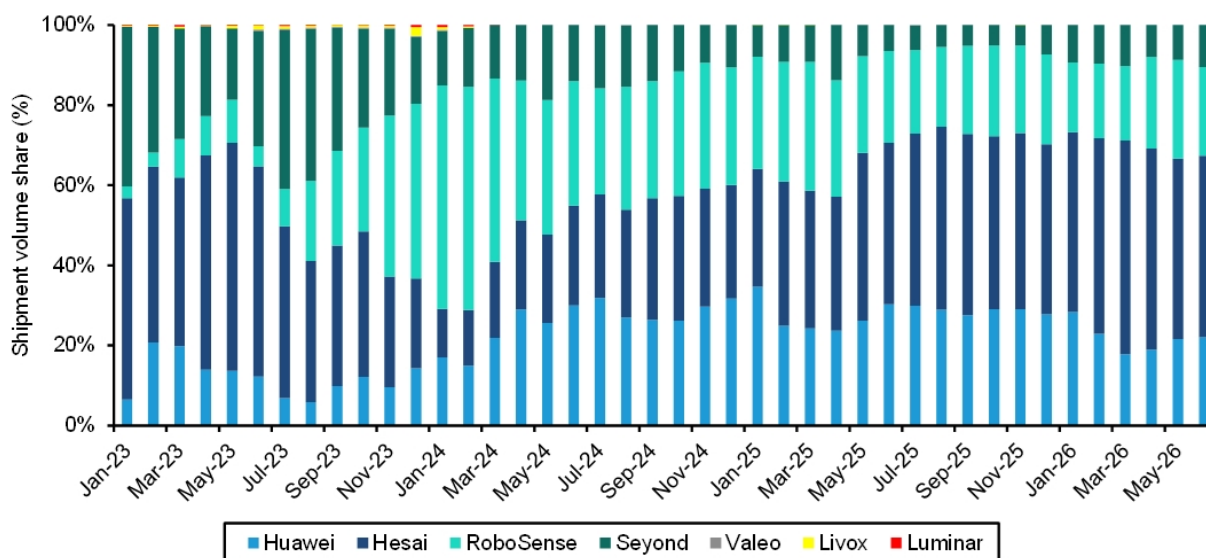
China: Long-range ADAS LiDAR shipment volume and penetration rate



Source: Gasgoo, Bernstein analysis

EXHIBIT 31: Long-range ADAS LiDAR in China: Market share breakdown

China: Long-range ADAS LiDAR shipment volume share



Source: Gasgoo, Bernstein analysis

APPENDIX

The SAE International (the Society of Automotive Engineers) [J3016 standard](#) categorized smart driving into 6 levels of automation, from L0 to L5 (Exhibit 5), which is commonly used as the standard categorization. Based on that definition, we first divide smart driving into two distinct stages: ADAS (Advanced Driver Assistance Systems) and AD (Autonomous Driving). In ADAS (from L0 to L2+/L2++), the human driver retains full responsibility for vehicle operation, even when assisted by automated features. Conversely, in AD (from L3 to Above), the system assumes responsibility for driving tasks under specific conditions. Since full autonomous driving (AD) is not yet legally approved for passenger vehicles (PV), we expect most PVs to remain at Level 2 (ADAS). However, the original Level 2 classification alone does not sufficiently capture the varying degrees of smart driving capabilities, especially with the introduction of the NOA (Navigate on Autopilot) features. **Therefore, we define a basic Level 2 without NOA functionality as L2, a mid-level Level 2 with Highway NOA as L2+, and an advanced Level 2 with Urban NOA as L2++ to capture the nuance in autonomous driving hardware capabilities.**

More details on our definition below:

- **L0** means the human driver performs all driving tasks, despite autonomous warning and support systems, like blind-spot alerts or emergency braking.
- **L1** is classified as the driver assistance, characterized as Feet Off. The human driver is always responsible, even if the system can assist with either steering or acceleration/braking.
- **L2** refers to partial automation, characterized as Hands Off in certain situations. Both steering and speed can be automated, but driver must monitor and stay ready. In our definition of L2, it only refers to basic level of L2 without NOA functionality.
- **L2+**, also called L2.5 in some cases, is defined as cars with capability up to Highway NOA but is not able to conduct Urban NOA functionality without pre-set route based training. Typical functions include automatic lane change, merging, exit handling, and route following in highway. In order to support the Highway NOA functionality, the smart driving chip requirement is significantly higher than basic L2 level.
- **L2++**, also called L2.9 in some cases, is defined as cars with Urban NOA capability, which can automatically drive in the majority of city routes without pre-train. L2++ are almost comparable to SAE L3 in terms of smart driving capability and thus the smart driving chip requirement, offering origin-to-destination autonomous navigation capabilities. Compared to Highway NOA (L2+) systems, Urban NOA (L2++) involves a significantly higher level of complexity for real-time calculation, representing a full order of magnitude increase in sophistication. Consequently, the demand for TOPS in smart driving chips is also several times greater, making it necessary to create another sub-segment on top of the L2+ segment.
- **L3** introduces conditional automation, characterized as Eyes Off. L3 allows the system to perform driving tasks under specific conditions, but still requiring the driver to take over when requested.
- **L4** represents high automation, characterized as Mind Off. No driver attention is needed in certain geo-fenced areas or scenarios, but human being is still required to sit on the driver's seat.
- **L5** is full automation, characterized as Driver Off. No human needed at all; system operates independently in any condition, anywhere.

Given the fewer regulatory hurdles and enhanced safety assurances, L2++ (Urban NOA) is anticipated to become the predominant solution in the future, offering a balanced approach between technological advancement and practical implementation.

EXHIBIT 32: **L2+ & L2++ in functionality is closed to L3 but is still defined as ADAS. The core distinction between L2+&L2++ and L3 lies in the shift of responsibility from drivers to system/OEM**

	ADAS (Advanced driver assistance systems) Driver takes responsibility					AD (Autonomous driving) Car takes responsibility		
	L 0	L 1	L 2	L2+ (Highway NOA)	L2++ (Urban NOA)	L 3	L 4	L 5
Definition description	Manual driving + Drivers fully control the vehicle, even though getting assistance from alert and protection system	Assisted driving + Auxiliary driving system (ADS) can control either speed or steering wheel + Other driving behaviors are done by drivers	Partial automated driving + ADS controls speed and steering wheel + Other driving behaviors are done by drivers + Requested lane-change possible	Temp. automated driving in highway + Self-initiated lane change, speed control, and adaptive cruise control in highway + Drivers step in based on system requests	Temp. automated driving in urban + Self-initiated lane change, speed control, and adaptive cruise control in city + Drivers step in based on system requests	Temp. automated driving + Autonomous driving system drives in defined ODDs + Drivers step in based on system requests	Fully autonomous -limited ODD + Autonomous driving system drives + Drivers don't have to respond + Limited ODD (roads and conditions)	Fully autonomous -all ODDs + Autonomous driving system drives + Drivers don't have to respond + Adapt all roads and conditions
Feet-off	X	✓	✓	✓	✓	✓	✓	✓
Hands-off	X	X	✓	✓	✓	✓	✓	✓
Eyes-off	X	X	X	X	X	✓	✓	✓
Mind-off	X	X	X	X	X	X	✓	✓
Driver-off	X	X	X	X	X	X	X	✓
Aid visibility & Warn	 Drivers			Major battleground		 System		
Execute driving								
Supervise driving								
Emergency step in								

Source: SEA, Bernstein analysis

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